

Solutions to

Daron Acemoglu

Introduction to Modern Economic Growth

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Chapters 2–3

The exercises that the instructor’s solutions manual of Peters and Simsek (2009) leaves unsolved: Exercises 2.1–2.6, 2.8–2.10, 2.13, 2.15, 2.24–2.26 and 3.3–3.8, 3.11, with the numerical parts computed by the Python programs in `code/`.

The exercise statements are not reproduced here: read them in Acemoglu’s *Introduction to Modern Economic Growth*.

Each solution is headed by the number of its exercise in the book; notation, the part letters (a), (b), ... follow the statements there.

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Chapter 2

The Solow Growth Model

Solutions to the fourteen exercises of Chapter 2 of Daron Acemoglu, *Introduction to Modern Economic Growth* (Princeton University Press, 2009), that are left unsolved in the instructor's solutions manual of Peters and Simsek (2009): Exercises 2.1–2.6, 2.8, 2.9, 2.10, 2.13, 2.15, 2.24, 2.25 and 2.26. The remaining thirteen exercises of the chapter (2.7, 2.11, 2.12, 2.14, 2.16–2.23 and 2.27) are solved there. Assumptions, propositions and equations cited as ASSUMPTION 2, PROPOSITION 2.5 or (2.18) refer to the book. Throughout, F is the aggregate production function, $f(k) \equiv F(k, 1, A)$ the per capita production function, $k \equiv K/L$ the capital-labor ratio, $w(k) \equiv f(k) - kf'(k)$ the wage, and $R(k) \equiv f'(k)$ the rental rate of capital. The numerical parts are computed by `code/ch2.py`.

Exercise 2.1

Fix a date t and write $\bar{L}(t) > 0$ for the (finite, inelastically supplied) labor endowment and $K(t)$ for the capital stock used in production. In a competitive equilibrium the representative firm solves (2.5),

$$\max_{K \geq 0, L \geq 0} F(K, L, A(t)) - R(t)K - w(t)L,$$

taking $(R(t), w(t))$ as given, and factor markets clear in the complementary slackness form (2.4):

$$L(t) \leq \bar{L}(t), \quad w(t) \geq 0, \quad (L(t) - \bar{L}(t))w(t) = 0.$$

Suppose, to get a contradiction, that $w(t) = 0$. Let $(K(t), L(t))$ be the equilibrium factor demands, which must be finite since they cannot exceed the endowments. Consider raising employment to $L(t) + \varepsilon$ for $\varepsilon > 0$, holding capital fixed. Profits change by

$$F(K(t), L(t) + \varepsilon, A(t)) - F(K(t), L(t), A(t)) - w(t)\varepsilon = \int_{L(t)}^{L(t)+\varepsilon} F_L(K(t), \ell, A(t)) d\ell > 0,$$

because $w(t) = 0$ and $F_L > 0$ everywhere by ASSUMPTION 1. So the firm strictly prefers more labor at *every* finite L : the maximization problem has no solution, and in particular $(K(t), L(t))$ is not profit maximizing. That contradicts firm optimization, so $w(t) > 0$.

Given $w(t) > 0$, the complementary slackness condition in (2.4) forces $L(t) - \bar{L}(t) = 0$, which is exactly the simple market clearing condition (2.3), $L(t) = \bar{L}(t)$. This is why (2.3) can be used throughout in place of (2.4).

Remark. Only $F_L > 0$ is used; neither constant returns nor the Inada conditions are needed. The identical argument with $F_K > 0$ shows $R(t) > 0$, so the capital market also clears with equality, $K(t) = \bar{K}(t)$. Note also that *some* bound on labor demand is essential to the argument: it is the finiteness of the endowment that turns “the firm always wants more labor” into a contradiction.

Exercise 2.2

Fix A and write $F(K, L)$ for $F(K, L, A)$, which by ASSUMPTION 1 is twice differentiable and homogeneous of degree 1 in (K, L) , with $F_{KK} < 0$ and $F_{LL} < 0$.

The Hessian is singular. By THEOREM 2.1 (Euler's Theorem), the partial derivatives F_K and F_L are homogeneous of degree 0. Applying Euler's Theorem to each of them,

$$KF_{KK} + LF_{KL} = 0 \quad \text{and} \quad KF_{LK} + LF_{LL} = 0.$$

Hence, for $K, L > 0$,

$$F_{KL} = -\frac{K}{L}F_{KK} > 0 \quad \text{and} \quad F_{KL} = -\frac{L}{K}F_{LL} > 0,$$

the signs following from $F_{KK} < 0$ and $F_{LL} < 0$. Multiplying the two expressions,

$$F_{KL}^2 = \left(-\frac{K}{L}F_{KK}\right)\left(-\frac{L}{K}F_{LL}\right) = F_{KK}F_{LL},$$

so that the Hessian

$$\mathbf{H} = \begin{pmatrix} F_{KK} & F_{KL} \\ F_{KL} & F_{LL} \end{pmatrix} \quad \text{has} \quad \det \mathbf{H} = F_{KK}F_{LL} - F_{KL}^2 = 0.$$

Concavity. A twice differentiable function is concave if and only if its Hessian is negative semidefinite everywhere. Here the leading principal minors are $F_{KK} < 0$ and $\det \mathbf{H} = 0$, so $\mathbf{H}$ is negative semidefinite (its eigenvalues are 0 and $F_{KK} + F_{LL} < 0$). Therefore F is concave in (K, L) .

But not strictly. Strict concavity would require $\mathbf{H}$ to be negative definite, which fails since $\det \mathbf{H} = 0$; the null direction is $(K, L)'$ itself, because $\mathbf{H} \begin{pmatrix} K \\ L \end{pmatrix} = \begin{pmatrix} KF_{KK} + LF_{KL} \\ KF_{LK} + LF_{LL} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$. The failure is easy to see directly: along any ray through the origin, constant returns make F linear,

$$F(\lambda K + (1 - \lambda)\lambda'K, \lambda L + (1 - \lambda)\lambda'L) = [\lambda + (1 - \lambda)\lambda']F(K, L) = \lambda F(K, L) + (1 - \lambda)F(\lambda'K, \lambda'L),$$

so the concavity inequality holds with equality for any two points on a ray. This is the sense in which the production function can be “concave but not strictly concave” even though $F_{KK} < 0$ and $F_{LL} < 0$: it curves in every direction except the direction in which the two inputs are scaled together.

Remark. The per capita function $f(k) \equiv F(k, 1)$ is strictly concave, since $f''(k) = F_{KK}(k, 1) < 0$. Strict concavity of f , not of F , is what the uniqueness arguments of the chapter use.

Exercise 2.3

Write the profit function as

$$\pi(K, L) \equiv F(K, L, A) - RK - wL.$$

Since F is homogeneous of degree 1 in (K, L) , so is π : for every $\lambda > 0$,

$$\pi(\lambda K, \lambda L) = \lambda \pi(K, L).$$

For $L > 0$ write $\kappa \equiv K/L$; then constant returns give

$$\pi(K, L) = L[f(\kappa) - R\kappa - w] \equiv L\varphi(\kappa), \quad f(\kappa) \equiv F(\kappa, 1, A).$$

Everything therefore depends on the sign of the maximized value of φ , and only three cases can arise.

Case 1: $\varphi(\kappa) > 0$ for some κ (equivalently $\pi(K_0, L_0) > 0$ for some (K_0, L_0)). Then $\pi(\lambda K_0, \lambda L_0) = \lambda\pi(K_0, L_0) \rightarrow \infty$ as $\lambda \rightarrow \infty$: profits are unbounded and no maximizer exists. This is the case in which factor prices are “too low” relative to marginal products.

Case 2: $\varphi(\kappa) < 0$ for every $\kappa \geq 0$. Then $\pi(K, L) < 0$ whenever $L > 0$. On the boundary $L = 0$ constant returns give $F(K, 0, A) = K F(1, 0, A)$, and

$$F(1, 0, A) = \lim_{L \rightarrow 0} Lf(1/L) = \lim_{k \rightarrow \infty} \frac{f(k)}{k} = 0$$

by ASSUMPTION 2, so $\pi(K, 0) = -RK \leq 0$ with equality only at $K = 0$. Since $\pi(0, 0) = 0$, the unique maximizer is $K = L = 0$ and maximal profit is zero.

Case 3: $\max_{\kappa} \varphi(\kappa) = 0$, attained at some $\kappa^* > 0$. Then $\pi(K, L) = 0$ for every (K, L) with $K/L = \kappa^*$, and $\pi(K, L) < 0$ otherwise (for $L > 0$), so the solution set is the entire ray

$$\{(K, L) : K/L = \kappa^*, L \geq 0\},$$

a continuum: the scale of the firm is completely indeterminate, only its capital-labor ratio is pinned down.

These are the only possibilities, since $\max_{\kappa} \varphi$ is either positive, negative, or zero. In equilibrium it is Case 3 that must obtain: factor supplies are fixed and finite, so Case 1 is inconsistent with market clearing and Case 2 would leave the supplies unemployed. Hence equilibrium factor prices satisfy $\varphi(\kappa^*) = 0$ with $\varphi'(\kappa^*) = f'(\kappa^*) - R = 0$, which is (2.7), and then $w = f(\kappa^*) - \kappa^* f'(\kappa^*)$, which is (2.6); profits are zero, as PROPOSITION 2.1 states. The indeterminacy of scale is why the model determines only aggregates, and why the representative firm can be taken to operate at the economy's endowment point.

Exercise 2.4

It is convenient to factor the function once and for all:

$$f(k) = k(k-1)(k-2)(k-3), \quad \text{so} \quad \frac{f(k)}{k} = (k-1)(k-2)(k-3).$$

Also

$$\begin{aligned} f'(k) &= 4k^3 - 18k^2 + 22k - 6, & f''(k) &= 12k^2 - 36k + 22, \\ w(k) &= f(k) - kf'(k) = -k^2(3k^2 - 12k + 11). \end{aligned}$$

(a) *Violations.* The underlying production function is $F(K, L) = Lf(K/L)$, so $F_K = f'(k)$, $F_{KK} = f''(k)/L$, $F_L = w(k)$ and $F_{LL} = (K^2/L^3)f''(k)$. Constant returns to scale hold by construction, and f is a polynomial, hence twice (indeed infinitely) differentiable. Everything else in ASSUMPTION 1 fails somewhere:

- *Nonnegativity of output.* $f(k) < 0$ for $k \in (0, 1) \cup (2, 3)$ — for instance $f(1/2) = -0.9375$ — so F does not map into $\mathbb{R}_+$.
- *Positive marginal products.* $f'(0) = -6 < 0$, so $F_K > 0$ fails near $k = 0$ (and again on part of $(2, 3)$). Likewise $w(k) < 0$ for $k < 2 - 1/\sqrt{3}$ and for $k > 2 + 1/\sqrt{3}$, e.g. $w(3) = -18$, so $F_L > 0$ fails.
- *Diminishing marginal products.* $f''(k) = 12k^2 - 36k + 22$ is positive outside the interval between its roots $k = (9 \pm \sqrt{15})/6$, i.e. on $(0, 0.8545) \cup (2.1455, \infty)$ — for instance $f''(0) = 22 > 0$. So $F_{KK} < 0$ and $F_{LL} < 0$ both fail there.

Of ASSUMPTION 2, only $F(0, L, A) = 0$ survives ($f(0) = 0$). Both Inada conditions for capital fail: $\lim_{k \rightarrow 0} f'(k) = -6 \neq \infty$ and $f'(k) \rightarrow +\infty$ rather than 0 as $k \rightarrow \infty$ (e.g. $f'(10) = 2414$). The Inada conditions for labor fail as well, since $F_L = w(k) \rightarrow -\infty$ as $k \rightarrow \infty$.

(b) *Three steady states.* With population growth n and depreciation δ , the fundamental law of motion (2.33) is

$$\frac{\dot{k}(t)}{k(t)} = s \frac{f(k(t))}{k(t)} - (n + \delta),$$

so an interior steady state is a solution of

$$(k-1)(k-2)(k-3) = h, \quad h \equiv \frac{n+\delta}{s} > 0. \quad (*)$$

The cubic on the left has derivative $3k^2 - 12k + 11$, which vanishes at $k = 2 \mp 1/\sqrt{3}$; it therefore rises to a local maximum at $k = 2 - 1/\sqrt{3} \approx 1.42265$, falls to a local minimum at $k = 2 + 1/\sqrt{3}$, and rises thereafter. The two extreme values are

$$\left(1 - \frac{1}{\sqrt{3}}\right)\left(-\frac{1}{\sqrt{3}}\right)\left(-1 - \frac{1}{\sqrt{3}}\right) = \frac{2}{3\sqrt{3}} = \frac{2\sqrt{3}}{9} \approx 0.38490 \quad \text{and} \quad -\frac{2\sqrt{3}}{9}.$$

Hence (*) has exactly three positive roots whenever

$$0 < \frac{n+\delta}{s} < \frac{2\sqrt{3}}{9} \approx 0.38490,$$

one in $(1, 2 - 1/\sqrt{3})$, one in $(2 - 1/\sqrt{3}, 2)$, and one in $(3, \infty)$; it has exactly one (in $(3, \infty)$) when $h > 2\sqrt{3}/9$. For example, with $n = 0$ and $\delta/s = 1/4$ the three steady states are

$$k_1^* = 1.162435, \quad k_2^* = 1.730406, \quad k_3^* = 3.107160.$$

Figure 1 plots the cubic together with the horizontal line h .

(c) *Stability.* Since $\dot{k} = k s [(k-1)(k-2)(k-3) - h]$, the sign of $\dot{k}$ is the sign of the vertical distance between the cubic and the line, and at a steady state

$$\left. \frac{\partial \dot{k}}{\partial k} \right|_{k=k^*} = s k^* \cdot \frac{d}{dk} \left[\frac{f(k)}{k} \right]_{k=k^*},$$

so a steady state is locally stable exactly when the cubic crosses the line *downward*. The cubic crosses upward at k_1^* (rising branch), downward at k_2^* (falling branch), and upward again at k_3^* :

$$\text{at } k_1^* : +1.1045, \quad \text{at } k_2^* : -0.7820, \quad \text{at } k_3^* : +2.6774,$$

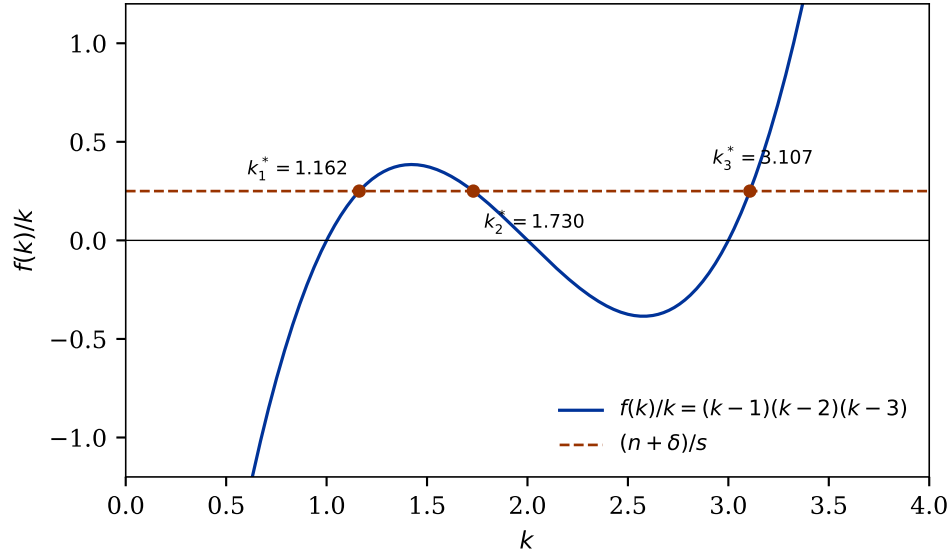


FIGURE 1. Steady states of the quartic economy: the interior steady states are the intersections of $f(k)/k = (k-1)(k-2)(k-3)$ with the horizontal line $(n+\delta)/s$, drawn here at $1/4$.

for the derivative of $f(k)/k$. Therefore k_2^* is locally asymptotically stable while k_1^* and k_3^* are unstable. In addition $k = 0$ is a rest point of the differential equation, and it is locally stable: for small $k > 0$ we have $f(k)/k < 0 < h$, so $\dot{k} < 0$ and the economy collapses to zero. The complete phase line is

$$\underbrace{0}_{\text{stable}} \longleftarrow \underbrace{k_1^*}_{\text{unstable}} \longrightarrow \underbrace{k_2^*}_{\text{stable}} \longleftarrow \underbrace{k_3^*}_{\text{unstable}} \longrightarrow \infty.$$

An economy starting with $k(0) < k_1^*$ collapses to $k = 0$; one starting in (k_1^*, k_3^*) converges to k_2^* ; one starting above k_3^* grows without bound. This is a poverty-trap configuration: history (the initial condition) decides the long-run outcome.

Finally, *no* steady state can be globally stable. Global asymptotic stability requires convergence to the same point from every initial condition; but an economy that starts at another steady state stays there forever, so the existence of more than one rest point rules global stability out immediately.

Remark. Part (c) of the exercise asks one to prove that *two* of the three steady states in (b) are locally stable. Of the three *interior* steady states exactly one — the middle one — is locally stable, as the sign pattern above shows: on the rising branches of $f(k)/k$ the economy is pushed away from the steady state. The count in the book is recovered only if the degenerate rest point $k = 0$, which the text elsewhere excludes as having “no economic content”, is included among the steady states; then two of the four rest points ($k = 0$ and k_2^*) are stable and two are unstable.

Exercise 2.5

We must show that under ASSUMPTIONS 1 and 2 the continuous-time Solow model with population growth at the rate n has a unique steady-state equilibrium, with $k^* \in (0, \infty)$ solving (2.34), $y^* = f(k^*)$ and $c^* = (1-s)f(k^*)$.

By (2.33) the equilibrium path satisfies $\dot{k}(t)/k(t) = sf(k(t))/k(t) - (n + \delta)$, so a steady state with $k > 0$ is precisely a solution of

$$\psi(k) \equiv \frac{f(k)}{k} = \frac{n + \delta}{s}. \quad (*)$$

ψ is continuous and strictly decreasing. Continuity is immediate from ASSUMPTION 1. For monotonicity, differentiate:

$$\psi'(k) = \frac{f'(k)k - f(k)}{k^2} = -\frac{w(k)}{k^2}, \quad w(k) = f(k) - kf'(k).$$

Since f is strictly concave with $f(0) = 0$ (ASSUMPTION 2), strict concavity gives $f(k) > f(0) + kf'(k) = kf'(k)$, that is $w(k) > 0$ for every $k > 0$; equivalently, $w(k) = F_L > 0$ is the wage, which is positive by ASSUMPTION 1. Hence $\psi'(k) < 0$ for all $k > 0$.

The limits. Both $f(k) \rightarrow 0$ and $k \rightarrow 0$ as $k \rightarrow 0$, so l'Hôpital's rule and ASSUMPTION 2 give

$$\lim_{k \rightarrow 0} \psi(k) = \lim_{k \rightarrow 0} f'(k) = \lim_{K \rightarrow 0} F_K = \infty, \quad \lim_{k \rightarrow \infty} \psi(k) = \lim_{k \rightarrow \infty} f'(k) = \lim_{K \rightarrow \infty} F_K = 0.$$

(The second limit also follows from $0 < f(k)/k$ and concavity.)

Existence and uniqueness. ψ is continuous on $(0, \infty)$ with limits ∞ at 0 and 0 at ∞ , so by the Intermediate Value Theorem it attains the value $(n + \delta)/s > 0$ at some $k^* \in (0, \infty)$; since ψ is strictly decreasing, that value is attained exactly once. Hence there is a unique steady-state capital-labor ratio, given by (2.34).

Finally, at k^* the level of output per capita is $y^* = f(k^*)$ by definition of f , and since a constant fraction s of income is saved, consumption per capita is $c^* = (1 - s)f(k^*)$, with the corresponding investment $sf(k^*) = (n + \delta)k^*$ exactly replacing depreciated capital and equipping the new workers.

Remark. The proof is that of PROPOSITION 2.2 with δ replaced by $n + \delta$: going from discrete to continuous time, and adding population growth, changes nothing essential. Population growth acts exactly like faster depreciation, since both erode the capital-labor ratio at a constant proportional rate.

Exercise 2.6

Let $f(k) = A\tilde{f}(k)$ with $\tilde{f}$ satisfying ASSUMPTIONS 1 and 2, and let k^* be the unique steady state of Exercise 2.5, i.e. the solution of

$$G(k^*; A, s, \delta, n) \equiv sA \frac{\tilde{f}(k^*)}{k^*} - (n + \delta) = 0.$$

Writing $\tilde{w}(k) \equiv \tilde{f}(k) - k\tilde{f}'(k) > 0$ for the wage per unit of A , the computation in Exercise 2.5 gives

$$\frac{\partial G}{\partial k^*} = sA \cdot \frac{d}{dk} \left[\frac{\tilde{f}(k)}{k} \right]_{k=k^*} = -\frac{sA \tilde{w}(k^*)}{(k^*)^2} < 0,$$

so the Implicit Function Theorem applies at every parameter configuration and yields $\partial k^*/\partial x = -(\partial G/\partial x)/(\partial G/\partial k^*)$ for each parameter x . Since $\partial G/\partial A = s\tilde{f}(k^*)/k^* > 0$, $\partial G/\partial s = A\tilde{f}(k^*)/k^* > 0$, $\partial G/\partial \delta = -1$ and $\partial G/\partial n = -1$,

$$\frac{\partial k^*}{\partial A} = \frac{k^* \tilde{f}(k^*)}{A \tilde{w}(k^*)} > 0, \quad \frac{\partial k^*}{\partial s} = \frac{k^* \tilde{f}(k^*)}{s \tilde{w}(k^*)} > 0, \quad \frac{\partial k^*}{\partial \delta} = \frac{\partial k^*}{\partial n} = -\frac{(k^*)^2}{sA \tilde{w}(k^*)} < 0.$$

This proves the four statements about k^* .

For output, $y^* = A\tilde{f}(k^*)$, so for $x \in \{s, \delta, n\}$ the chain rule gives

$$\frac{\partial y^*}{\partial x} = A\tilde{f}'(k^*)\frac{\partial k^*}{\partial x},$$

which has the same sign as $\partial k^*/\partial x$ because $\tilde{f}' > 0$: $\partial y^*/\partial s > 0$ and $\partial y^*/\partial \delta, \partial y^*/\partial n < 0$. For A there is in addition a direct effect, and both effects push the same way:

$$\frac{\partial y^*}{\partial A} = \tilde{f}(k^*) + A\tilde{f}'(k^*)\frac{\partial k^*}{\partial A} > 0.$$

The economics is transparent. A higher saving rate or a better technology raises investment at every capital-labor ratio and so shifts the $sf(k)/k$ schedule up, moving its intersection with the horizontal line $n + \delta$ to the right. A higher depreciation rate or a higher rate of population growth raises the amount of investment needed merely to hold k constant — capital must be replaced, and new workers must be equipped — and so moves the intersection to the left. The only new result relative to PROPOSITION 2.3 is the effect of n , and it enters exactly as δ does: countries with faster population growth are, other things equal, poorer in per capita terms.

Exercise 2.8

Throughout, F is twice differentiable and exhibits constant returns to scale, $f(k) = F(k, 1, A)$ is concave (so $f'' \leq 0$) with $f(0) = 0$, and the Inada conditions of ASSUMPTION 2 hold. What is *not* assumed is $F_{KK} < 0$, $F_{LL} < 0$, or even $F_K > 0$ and $F_L > 0$. The first thing to notice is that the missing sign restrictions on the first derivatives come for free.

Step 1: the marginal product of capital is nonnegative. Concavity makes f' nonincreasing. If $f'(k_0) < 0$ for some k_0 , then $f'(k) \leq f'(k_0) < 0$ for every $k \geq k_0$, contradicting the Inada condition $\lim_{k \rightarrow \infty} f'(k) = 0$. Hence $f'(k) \geq 0$, that is $F_K \geq 0$, for all $k > 0$. (One cannot get more: a technology that is affine from some k_0 on has $f' \equiv 0$ beyond k_0 and still satisfies every hypothesis. As the rest of the argument shows, $f' \geq 0$ is all that is needed.)

Step 2: the wage is strictly positive. Write $w(k) = f(k) - kf'(k)$. Since $f(0) = 0$,

$$w(k) = \int_0^k [f'(x) - f'(k)] dx,$$

and the integrand is nonnegative because f' is nonincreasing. It is moreover strictly positive on a set of positive measure: $f'(x) \rightarrow \infty$ as $x \rightarrow 0$ while $f'(k) < \infty$, so $f'(x) > f'(k)$ for all small x . Hence $w(k) > 0$ for every $k > 0$; that is, $F_L > 0$, and the “not necessarily strictly” concave technology still pays a positive wage.

PROPOSITION 2.2: *existence and uniqueness of the steady state.* Exactly as in Exercise 2.5, $\psi(k) \equiv f(k)/k$ satisfies $\psi'(k) = -w(k)/k^2 < 0$ by Step 2, so ψ is *strictly* decreasing even though f is only weakly concave. Its limits are ∞ at 0 and 0 at ∞ by ASSUMPTION 2 and l'Hôpital's rule. The Intermediate Value Theorem therefore gives a unique $k^* \in (0, \infty)$ with $f(k^*)/k^* = \delta/s$, and then $y^* = f(k^*)$ and $c^* = (1-s)f(k^*)$. Nothing in the argument used strict concavity; it used only that $w > 0$, which Step 2 delivers.

PROPOSITION 2.5: *global stability.* Let $g(k) \equiv sf(k) + (1-\delta)k$, so that the discrete time law of motion is $k(t+1) = g(k(t))$ and $k^* = g(k^*)$. By Step 1, $g'(k) = sf'(k) + 1 - \delta > 0$. By Step 2,

$\delta = sf(k^*)/k^* > sf'(k^*)$, whence

$$g'(k^*) = sf'(k^*) + 1 - \delta \in (0, 1),$$

and local stability follows from COROLLARY 2.1. For global stability, take $k(t) \in (0, k^*)$. Since g is strictly increasing, $k(t+1) = g(k(t)) < g(k^*) = k^*$; and since ψ is strictly decreasing,

$$\frac{k(t+1) - k(t)}{k(t)} = s \frac{f(k(t))}{k(t)} - \delta > s \frac{f(k^*)}{k^*} - \delta = 0.$$

So $k(t+1) \in (k(t), k^*)$: the sequence is increasing and bounded above by k^* , hence convergent, and its limit is a fixed point of g in $(0, k^*]$, which can only be k^* by uniqueness. The argument starting from $k(0) > k^*$ is identical with the inequalities reversed. Monotone global convergence therefore survives the weakening of ASSUMPTION 1.

PROPOSITION 2.6 *must be weakened to weak monotonicity*. The proposition asserts that $\{w(t)\}$ is an increasing and $\{R(t)\}$ a decreasing sequence when $k(0) < k^*$. With $f'' \leq 0$ instead of $f'' < 0$ we get

$$R'(k) = f''(k) \leq 0 \quad \text{and} \quad w'(k) = -kf''(k) \geq 0,$$

so the correct statement is that $\{w(t)\}$ is *nondecreasing* and $\{R(t)\}$ is *nonincreasing* (with the inequalities reversed when $k(0) > k^*$). Both can be locally constant: on any interval where f is affine — the production function is then locally an AK technology — the capital-labor ratio keeps rising while factor prices do not move at all. Strict monotonicity is restored as soon as $f'' < 0$ on the relevant range, which is what ASSUMPTION 1 guarantees.

Exercise 2.9

Let ASSUMPTIONS 1 and 2 hold and suppose $k(0) < k^*$. By PROPOSITION 2.5 the capital-labor ratio increases monotonically to k^* : $k(t+1) > k(t)$ for every t , with $k(t) \rightarrow k^*$.

In equilibrium factor prices are the marginal products, and with constant returns they depend on the capital-labor ratio alone:

$$R(t) = F_K(K(t), L(t), A) = f'(k(t)),$$

$$w(t) = F_L(K(t), L(t), A) = f(k(t)) - k(t)f'(k(t)) = w(k(t)).$$

Differentiating the two expressions in k ,

$$\frac{dR}{dk} = f''(k) < 0, \quad \frac{dw}{dk} = f'(k) - f'(k) - kf''(k) = -kf''(k) > 0,$$

both strict by ASSUMPTION 1, which has $f''(k) = F_{KK}(k, 1, A) < 0$.

So $R(\cdot)$ is a strictly decreasing and $w(\cdot)$ a strictly increasing function of k . Composing with the strictly increasing sequence $\{k(t)\}$ gives

$$w(t+1) = w(k(t+1)) > w(k(t)) = w(t), \quad R(t+1) = f'(k(t+1)) < f'(k(t)) = R(t),$$

that is, $\{w(t)\}_{t=0}^\infty$ is an increasing and $\{R(t)\}_{t=0}^\infty$ a decreasing sequence, and they converge to $w(k^*)$ and $f'(k^*)$. If instead $k(0) > k^*$, PROPOSITION 2.5 gives a strictly decreasing $\{k(t)\}$ and the two conclusions are reversed.

The economics is the usual scarcity argument: capital deepening makes capital abundant relative to labor, so the marginal product of capital — and with it the rental rate and the interest rate $r = R - \delta$ — falls, while the marginal product of labor, and hence the wage, rises. Along the transition the entire increase in output per worker that is not saved accrues to labor.

Exercise 2.10

Part 1. The linear equation $\dot{x}(t) = ax(t)$ with $x(0)$ given has the unique solution $x(t) = x(0) \exp(at)$. If $a < 0$ then $\exp(at) \rightarrow 0$, so $x(t) \rightarrow 0$ for every initial condition: the steady state $x^* = 0$ is globally asymptotically stable. (If $a > 0$ the solution diverges unless $x(0) = 0$, and if $a = 0$ every point is a steady state.)

Part 2. Let g be differentiable in a neighborhood of x^* , with $g(x^*) = 0$ and $g'(x^*) < 0$. This is the scalar case ($n = 1$) of THEOREM 2.5: the Jacobian is the number $g'(x^*)$, its unique eigenvalue is $g'(x^*) < 0$, which has negative real part, so x^* is locally asymptotically stable. The direct argument is just as short. Since g is continuous with $g(x^*) = 0$ and $g'(x^*) < 0$, there is $\varepsilon > 0$ with

$$g(x) < 0 \text{ for } x \in (x^*, x^* + \varepsilon) \quad \text{and} \quad g(x) > 0 \text{ for } x \in (x^* - \varepsilon, x^*),$$

so on that neighborhood $\dot{x}$ always points toward x^* ; Part 3 applied to the interval $(x^* - \varepsilon, x^* + \varepsilon)$ then gives convergence.

Part 3. Let g be continuously differentiable with $g(x^*) = 0$, $g(x) < 0$ for $x > x^*$ and $g(x) > 0$ for $x < x^*$. Take any $x(0) < x^*$ (the case $x(0) > x^*$ is symmetric, and $x(0) = x^*$ is trivial).

Because g is continuously differentiable it is locally Lipschitz, so by the Picard–Lindelöf theorem the solution through any point is unique. The constant function $x \equiv x^*$ solves the equation, so the trajectory starting below x^* can never reach x^* in finite time, and *a fortiori* never crosses it: $x(t) < x^*$ for all t . Consequently $g(x(t)) > 0$ and $\dot{x}(t) > 0$ for all t , so $x(\cdot)$ is strictly increasing and bounded above by x^* . A monotone bounded function converges; let $L \equiv \lim_{t \rightarrow \infty} x(t) \leq x^*$.

Suppose $L < x^*$. By continuity $\dot{x}(t) = g(x(t)) \rightarrow g(L) > 0$, so there are T and $\eta > 0$ with $\dot{x}(t) \geq \eta$ for all $t \geq T$, whence $x(t) \geq x(T) + \eta(t - T) \rightarrow \infty$, contradicting $x(t) \leq x^*$. Therefore $L = x^*$, and $x(t) \rightarrow x^*$ from every initial condition: x^* is globally asymptotically stable.

Remark. Part 3 is what makes the continuous-time Solow model so much better behaved than its discrete-time counterpart. In discrete time a one-dimensional map can jump over its fixed point, so a sign condition on g alone does not preclude two-cycles; EXERCISE 2.21 constructs exactly such a cycle. In continuous time trajectories are continuous and cannot cross a rest point, which is why the analogue of Part 3 fails in discrete time.

Exercise 2.13

The law of motion. Labor income per worker is the wage $w(k) = f(k) - kf'(k)$, of which a fraction s is saved; capital income $f'(k)k$ is entirely consumed. Aggregate saving is therefore $sw(k(t))L(t)$, and with population growth at the rate n and depreciation δ the capital-labor ratio obeys

$$\dot{k}(t) = sw(k(t)) - (n + \delta)k(t), \quad \text{so} \quad \frac{\dot{k}(t)}{k(t)} = s\phi(k(t)) - (n + \delta), \quad \phi(k) \equiv \frac{w(k)}{k}.$$

Interior steady states are the solutions of

$$\phi(k) = \frac{w(k)}{k} = \frac{n + \delta}{s}. \quad (*)$$

The contrast with the baseline model is exactly the replacement of $f(k)/k$ by $w(k)/k$. In the baseline model $f(k)/k$ is strictly decreasing, which is what delivered uniqueness in Exercise 2.5. Nothing of the sort holds for $w(k)/k$.

When is ϕ increasing? Differentiating,

$$\phi'(k) = \frac{k w'(k) - w(k)}{k^2} = \frac{-k^2 f''(k) - w(k)}{k^2},$$

using $w'(k) = -k f''(k)$. Now recall that for a constant returns technology the elasticity of substitution (2.37) can be written as

$$\sigma(k) = \frac{f'(k) w(k)}{-k f(k) f''(k)},$$

so that $-k^2 f''(k) = \alpha_K(k) w(k)/\sigma(k)$ with $\alpha_K(k) \equiv k f'(k)/f(k)$ the share of capital in national income. Substituting,

$$\phi'(k) = \frac{w(k)}{k^2} \left[\frac{\alpha_K(k)}{\sigma(k)} - 1 \right], \quad \text{hence} \quad \phi'(k) > 0 \iff \sigma(k) < \alpha_K(k).$$

So $w(k)/k$ rises precisely where capital and labor are hard to substitute relative to the capital share. Intuitively, when σ is small an increase in capital raises the wage more than proportionately, because labor becomes the scarce complementary factor; saving out of wages then feeds back into more capital, and the economy can have several rest points.

An explicit example. Let

$$f(k) = A[\gamma k^\rho + (1 - \gamma)]^{1/\rho} + Bk^\alpha, \quad \rho = \frac{\sigma - 1}{\sigma} < 0, \quad \alpha \in (0, 1), \quad B > 0 :$$

a CES technology with a low elasticity of substitution plus a small Cobb–Douglas term. Both terms are increasing, strictly concave and homogeneous of degree one in (K, L) when written as $Lf(K/L)$, so ASSUMPTION 1 holds. The CES term alone would violate ASSUMPTION 2, since $f'_{\text{CES}}(0) = A\gamma^{1/\rho} < \infty$ and $w_{\text{CES}}(\infty) = A(1 - \gamma)^{1/\rho} < \infty$; the Cobb–Douglas term repairs both, because Bk^α contributes $B\alpha k^{\alpha-1} \rightarrow \infty$ to $f'(k)$ as $k \rightarrow 0$ and $(1 - \alpha)Bk^\alpha \rightarrow \infty$ to $w(k)$ as $k \rightarrow \infty$. So f satisfies ASSUMPTIONS 1 and 2 in full, and

$$w(k) = A(1 - \gamma)[\gamma k^\rho + (1 - \gamma)]^{(1-\rho)/\rho} + (1 - \alpha)Bk^\alpha.$$

Take $A = 1$, $\gamma = 1/2$, $\sigma = 1/4$ (so $\rho = -3$), $B = 0.02$ and $\alpha = 1/2$. Then $\phi(k) = w(k)/k$ falls from $+\infty$ to a local minimum $\phi = 0.03005$ at $k = 0.1509$, rises to a local maximum $\phi = 0.60360$ at $k = 1.4340$, and falls back to 0; the turning points are exactly where $\sigma(k)$ crosses $\alpha_K(k)$:

k	0.05	0.30	1.00	3.00
$\sigma(k)$	1.8871	0.3581	0.2537	0.3264
$\alpha_K(k)$	0.9667	0.9603	0.5000	0.0483
$\phi'(k)$	-0.438	+0.286	+0.495	-0.115

Consequently (*) has *three* solutions for every $(n + \delta)/s$ strictly between 0.03005 and 0.60360. With $(n + \delta)/s = 0.045$ they are

$$k_1^* = 0.04972, \quad k_2^* = 0.27684, \quad k_3^* = 29.19752,$$

and since $\dot{k}$ has the sign of $\phi(k) - (n + \delta)/s$, the outer two are locally asymptotically stable ($\phi'(k_1^*) = -0.442$, $\phi'(k_3^*) = -0.002$) while the middle one is unstable ($\phi'(k_2^*) = +0.240$). Figure 2 plots ϕ and the horizontal line. The economy converges to the poor or to the rich steady state according to whether it starts below or above k_2^* : with saving out of wages alone, the Solow model produces a poverty trap.

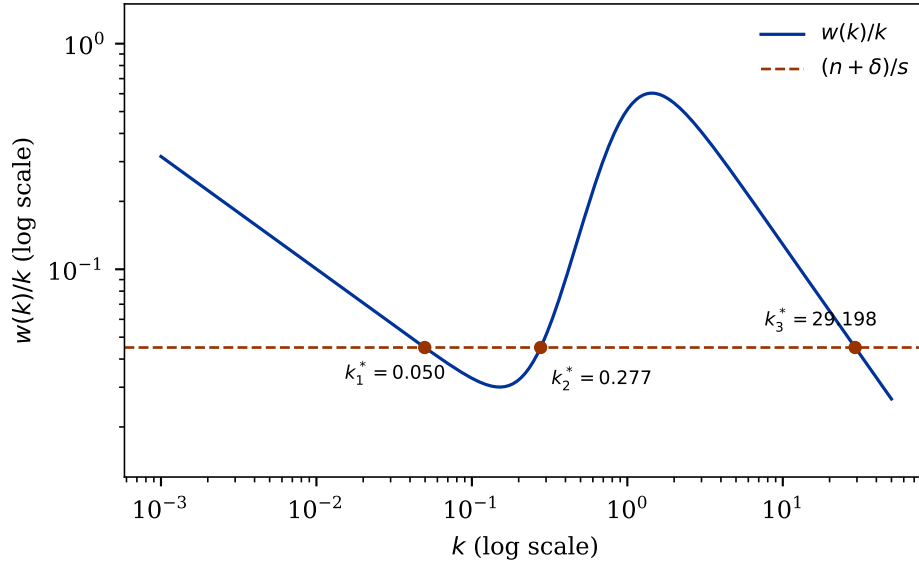


FIGURE 2. Saving out of labor income only: steady states are the intersections of $w(k)/k$ with the horizontal line $(n + \delta)/s$, drawn here at 0.045, for the technology of the text.

Remark. The comparison with EXERCISE 2.12 is instructive. When only capital income is saved, the law of motion is $\dot{k}/k = s_K f'(k) - (n + \delta)$, and f' is strictly decreasing by ASSUMPTION 1: the steady state is unique and globally stable, whatever the elasticity of substitution. It is the wage — the factor price that *rises* with capital — that can generate the self-reinforcing accumulation behind multiplicity.

Exercise 2.15

With constant returns, output per worker and the wage are both functions of the capital-labor ratio alone:

$$y = f(k), \quad w = F_L = f(k) - k f'(k),$$

the second equality being the competitive wage. Along the family of economies indexed by k ,

$$\frac{dy}{dk} = f'(k), \quad \frac{dw}{dk} = f'(k) - f'(k) - k f''(k) = -k f''(k) > 0,$$

so w is a strictly increasing function of k and can be inverted; by the chain rule,

$$\frac{\partial y}{\partial w} = \frac{dy/dk}{dw/dk} = \frac{f'(k)}{-k f''(k)}.$$

Therefore

$$\varepsilon_{y,w} = \frac{\partial y}{\partial w} \cdot \frac{w}{y} = \frac{f'(k)}{-kf''(k)} \cdot \frac{f(k) - kf'(k)}{f(k)} = \frac{f'(k)[f(k) - kf'(k)]}{-kf(k)f''(k)}. \quad (*)$$

It remains to check that the right-hand side of (*) is the elasticity of substitution (2.37). With constant returns, $F_K = f'(k)$ and $F_L = f(k) - kf'(k)$, so

$$\frac{\partial \log(F_K/F_L)}{\partial \log k} = k \frac{d}{dk} \left[\log f'(k) - \log(f(k) - kf'(k)) \right] = k \left[\frac{f''(k)}{f'(k)} + \frac{kf''(k)}{f(k) - kf'(k)} \right].$$

Putting the bracket over a common denominator,

$$\frac{\partial \log(F_K/F_L)}{\partial \log k} = kf''(k) \frac{[f(k) - kf'(k)] + kf'(k)}{f'(k)[f(k) - kf'(k)]} = \frac{kf(k)f''(k)}{f'(k)[f(k) - kf'(k)]},$$

and hence, by the definition (2.37),

$$\sigma = - \left[\frac{\partial \log(F_K/F_L)}{\partial \log(K/L)} \right]^{-1} = \frac{f'(k)[f(k) - kf'(k)]}{-kf(k)f''(k)},$$

which is exactly (*). Therefore $\varepsilon_{y,w} = \sigma$.

The result is the basis of the empirical strategy of Arrow et al. (1961) taken up in EXERCISE 2.16: regressing the logarithm of output per capita on the logarithm of the wage across economies (or across time) estimates the elasticity of substitution directly, and a slope of 1 — which is what those authors found to be a good approximation — says that the technology is Cobb–Douglas. It also gives a clean reading of σ : it measures how much richer, in output per worker, an economy must be in order to sustain a given proportional increase in its wage.

Exercise 2.24

Normalize the technology terms to $A_H = A_K = A_L = 1$ (they only rescale the arguments and the level, and none of the limits below depends on them) and write the per capita form of (2.38) as

$$f(k) = [\gamma k^\rho + (1 - \gamma)]^{1/\rho}, \quad \rho \equiv \frac{\sigma - 1}{\sigma} \in (-\infty, 1), \quad \gamma \in (0, 1),$$

so that $\sigma > 1 \iff \rho \in (0, 1)$, $\sigma < 1 \iff \rho < 0$ and $\sigma = 1$ is the Cobb–Douglas limit $f(k) = k^\gamma$. Differentiating,

$$f'(k) = \gamma k^{\rho-1} [\gamma k^\rho + (1 - \gamma)]^{(1-\rho)/\rho}, \quad w(k) = f(k) - kf'(k) = (1 - \gamma) [\gamma k^\rho + (1 - \gamma)]^{(1-\rho)/\rho}.$$

Recall that ASSUMPTION 2 requires four limits — $F_K \rightarrow \infty$ as $K \rightarrow 0$, $F_K \rightarrow 0$ as $K \rightarrow \infty$, $F_L \rightarrow \infty$ as $L \rightarrow 0$, $F_L \rightarrow 0$ as $L \rightarrow \infty$ — together with $F(0, L, A) = 0$. In per capita terms $F_K = f'(k)$ and $F_L = w(k)$, and $L \rightarrow 0$ with K fixed means $k \rightarrow \infty$, while $L \rightarrow \infty$ means $k \rightarrow 0$.

Case $\sigma > 1$ (i.e. $0 < \rho < 1$). As $k \rightarrow 0$ the bracket tends to $1 - \gamma > 0$, so

$$f(0) = (1 - \gamma)^{1/\rho} > 0,$$

and capital is not essential: $F(0, L, A) > 0$, the last requirement of ASSUMPTION 2, fails. As $k \rightarrow \infty$ the bracket behaves like γk^ρ , so

$$f'(k) \longrightarrow \gamma k^{\rho-1} (\gamma k^\rho)^{(1-\rho)/\rho} = \gamma^{1/\rho} > 0,$$

so $\lim_{K \rightarrow \infty} F_K = 0$ fails too: the marginal product of capital is bounded below by a positive number, and the technology is asymptotically linear in capital — an AK technology in the limit, which is exactly why EXERCISE 2.23(a) finds sustained growth in this case. Finally, as $k \rightarrow 0$ (that is, $L \rightarrow \infty$ for given K), the bracket tends to $1 - \gamma$ and

$$w(k) \rightarrow (1 - \gamma)(1 - \gamma)^{(1-\rho)/\rho} = (1 - \gamma)^{1/\rho} > 0,$$

so $\lim_{L \rightarrow \infty} F_L = 0$ fails as well: labor remains productive at the margin however abundant it is, because output does not require capital. (The two remaining limits, $f'(0) = \infty$ and $w(\infty) = \infty$, do hold, the latter because $w(k) \sim (1 - \gamma)\gamma^{(1-\rho)/\rho}k^{1-\rho}$ with $1 - \rho > 0$.)

Case $\sigma < 1$ (i.e. $\rho < 0$). Now $k^\rho \rightarrow \infty$ as $k \rightarrow 0$, so the bracket diverges and, since $1/\rho < 0$, $f(0) = 0$: capital is essential, as ASSUMPTION 2 wants. But

$$f'(k) \rightarrow \gamma k^{\rho-1} (\gamma k^\rho)^{(1-\rho)/\rho} = \gamma^{1/\rho} < \infty \quad \text{as } k \rightarrow 0,$$

so the Inada condition $\lim_{K \rightarrow 0} F_K = \infty$ fails. At the other end, as $k \rightarrow \infty$ the bracket tends to $1 - \gamma$ and

$$w(k) \rightarrow (1 - \gamma)^{1/\rho} < \infty,$$

so $\lim_{L \rightarrow 0} F_L = \infty$ fails as well. (The two remaining limits, $f'(\infty) = 0$ and $w(0) = 0$, do hold.)

Case $\sigma = 1$. Then $f(k) = k^\gamma$ with $\gamma \in (0, 1)$: $f(0) = 0$, $f'(k) = \gamma k^{\gamma-1} \rightarrow \infty$ as $k \rightarrow 0$ and $\rightarrow 0$ as $k \rightarrow \infty$, and $w(k) = (1 - \gamma)k^\gamma \rightarrow 0$ as $k \rightarrow 0$ and $\rightarrow \infty$ as $k \rightarrow \infty$. All the requirements of ASSUMPTION 2 hold.

Hence ASSUMPTION 2 holds for the CES technology if and only if $\sigma = 1$. The failure is economically meaningful rather than technical: with $\sigma > 1$ the two factors are good enough substitutes that capital alone can produce, and accumulating it never drives its marginal product to zero; with $\sigma < 1$ they are such poor substitutes that the first units of capital are not infinitely productive, and an economy with a fixed labor force cannot make the wage arbitrarily large.

Remark. ASSUMPTION 1, by contrast, holds for every finite $\sigma > 0$. Logarithmic differentiation of f' gives

$$\frac{f''(k)}{f'(k)} = \frac{\rho - 1}{k} + \frac{(1 - \rho)\gamma k^{\rho-1}}{\gamma k^\rho + (1 - \gamma)} = -\frac{(1 - \rho)(1 - \gamma)}{k[\gamma k^\rho + (1 - \gamma)]} < 0 \quad \text{for all } \rho < 1,$$

so $f'' < 0$, and $F_{KK}, F_{LL} < 0$ follow. The text on page 55 states instead that “the CES production function with $\sigma > 1$ violates Assumption 1”; what the computation above shows, and what this exercise asks one to prove, is that what fails is ASSUMPTION 2.

Exercise 2.25

With Harrod-neutral technological progress at the rate g and population growth at the rate n , the effective capital-labor ratio $k(t) = K(t)/(A(t)L(t))$ obeys (2.51), and the BGP value k^* solves (2.52),

$$H(k^*; s, \delta, n, g) \equiv s \frac{f(k^*)}{k^*} - (\delta + g + n) = 0.$$

The level of technology does not enter. $A(0)$ appears nowhere in this equation — it has been normalized away by the definition of k — so

$$\frac{\partial k^*(A(0), s, \delta, n, g)}{\partial A(0)} = 0.$$

The other comparative statics. Exactly as in Exercise 2.5,

$$\frac{\partial H}{\partial k^*} = s \cdot \frac{d}{dk} \left[\frac{f(k)}{k} \right]_{k=k^*} = -\frac{s w(k^*)}{(k^*)^2} < 0,$$

where $w(k) = f(k) - k f'(k) > 0$. The Implicit Function Theorem applies and gives, with $\partial H / \partial s = f(k^*) / k^* > 0$ and $\partial H / \partial \delta = \partial H / \partial n = \partial H / \partial g = -1$,

$$\frac{\partial k^*}{\partial s} = \frac{k^* f(k^*)}{s w(k^*)} > 0, \quad \frac{\partial k^*}{\partial n} = \frac{\partial k^*}{\partial \delta} = \frac{\partial k^*}{\partial g} = -\frac{(k^*)^2}{s w(k^*)} < 0.$$

Output per capita. Along the BGP, income per capita is $y^*(t) = A(t)f(k^*) = A(0)e^{gt}f(k^*)$ by (2.50). Hence, for each t ,

$$\frac{\partial y^*}{\partial A(0)} = e^{gt}f(k^*) > 0,$$

which is the one comparative static that is new relative to PROPOSITION 2.8: a higher initial technology level leaves the effective capital-labor ratio untouched but raises income per capita proportionately, at every date. For the remaining parameters the effect runs through k^* alone,

$$\frac{\partial y^*}{\partial x} = A(0)e^{gt}f'(k^*)\frac{\partial k^*}{\partial x}, \quad x \in \{s, \delta, n\},$$

which is positive for s and negative for δ and n , since $f' > 0$.

Remark. The proposition lists no comparative static with respect to g , and for good reason: a higher g lowers k^* and hence the level of income per capita at any fixed date, but it raises its growth rate, so $\partial y^*(t) / \partial g$ changes sign as t grows. This is the first appearance of a theme that recurs throughout the book — level effects and growth effects need not point the same way.

Exercise 2.26

Let ASSUMPTIONS 1 and 2 hold and consider the law of motion (2.51) of the effective capital-labor ratio,

$$\dot{k}(t) = G(k(t)), \quad G(k) \equiv sf(k) - (\delta + g + n)k.$$

G is continuously differentiable on $(0, \infty)$ by ASSUMPTION 1, and by PROPOSITION 2.11 it has a unique zero $k^* \in (0, \infty)$, characterized by $f(k^*)/k^* = (\delta + g + n)/s$.

Write $G(k) = k[s\psi(k) - (\delta + g + n)]$ with $\psi(k) \equiv f(k)/k$. As established in Exercise 2.5, ψ is strictly decreasing, with $\psi(k) \rightarrow \infty$ as $k \rightarrow 0$ and $\psi(k) \rightarrow 0$ as $k \rightarrow \infty$. Therefore

$$k < k^* \implies \psi(k) > \psi(k^*) = \frac{\delta + g + n}{s} \implies G(k) > 0,$$

$$k > k^* \implies \psi(k) < \psi(k^*) \implies G(k) < 0.$$

That is exactly the hypothesis of part 3 of COROLLARY 2.2 (proved in Exercise 2.10), which therefore gives global asymptotic stability: starting from any $k(0) > 0$, $k(t) \rightarrow k^*$. Moreover the convergence is monotone, since $\dot{k}$ keeps a constant sign along the path: an economy that starts

below its BGP accumulates effective capital throughout the transition, and one that starts above decumulates it.

Local stability can also be read off the derivative at the rest point. Using $sf(k^*) = (\delta + g + n)k^*$,

$$G'(k^*) = sf'(k^*) - (\delta + g + n) = sf'(k^*) - \frac{sf(k^*)}{k^*} = -\frac{sw(k^*)}{k^*} < 0,$$

because the wage $w(k^*) = f(k^*) - k^*f'(k^*)$ is strictly positive. So the linearization has a strictly negative eigenvalue, confirming local asymptotic stability by part 2 of COROLLARY 2.2, and giving in addition the speed of convergence: near the BGP the gap $k(t) - k^*$ closes at the exponential rate $sw(k^*)/k^* = (\delta + g + n)\alpha_L(k^*)$, where $\alpha_L = w/f$ is the share of labor in national income.

Since $k(t) \rightarrow k^*$ and $y(t) = A(t)f(k(t))$, output per capita converges to the balanced growth path $A(0)e^{gt}f(k^*)$ and grows asymptotically at the rate g : with Harrod-neutral progress, transitional dynamics and long-run growth separate cleanly, and the model behaves exactly as it did without technological change once variables are measured per efficiency unit of labor.

Chapter 3

The Solow Model and the Data

The seven exercises of Chapter 3 left unsolved in the instructor's manual: Exercises 3.3–3.8 and 3.11. The manual solves 3.1, 3.2, 3.9 and 3.10. Notation follows the book: $k \equiv K/(AL)$ is the effective capital-labor ratio, $\hat{y} \equiv Y/(AL)$, $y \equiv Y/L$, $y^*(t) \equiv A(t)f(k^*)$ is the steady-state (BGP) level of output per capita, and $\varepsilon_k(k) \equiv f'(k)k/f(k)$ is the elasticity of the per capita production function (3.9). The numerical parts are computed by `code/ch3.py`.

Exercise 3.3

Write the technology in labor-augmenting form, $Y = K^\alpha(AL)^{1-\alpha}$, so that in terms of the effective capital-labor ratio $k = K/(AL)$,

$$\hat{y} = k^\alpha, \quad y = Ak^\alpha, \quad \frac{\dot{k}}{k} = sk^{\alpha-1} - (\delta + g + n), \quad k^* = \left(\frac{s}{\delta + g + n} \right)^{\frac{1}{1-\alpha}}.$$

The exact equation. Differentiating $y = Ak^\alpha$ logarithmically and substituting the law of motion, with no approximation at all,

$$\frac{\dot{y}}{y} = g + \alpha \frac{\dot{k}}{k} = g + \alpha [sk^{\alpha-1} - (\delta + g + n)] = g + \alpha(\delta + g + n) \left[\left(\frac{k}{k^*} \right)^{\alpha-1} - 1 \right],$$

the last step using $s(k^*)^{\alpha-1} = \delta + g + n$. Since $y/y^* = (k/k^*)^\alpha$, we have $\log k - \log k^* = \alpha^{-1}(\log y - \log y^*)$, and therefore

$$\boxed{\frac{\dot{y}(t)}{y(t)} = g + \alpha(\delta + g + n) \left[\exp\left(-\frac{1-\alpha}{\alpha} (\log y(t) - \log y^*(t)) \right) - 1 \right]} \quad (*)$$

which is the exact analogue of (3.10). Expanding the exponential to first order around $\log y = \log y^*$ returns $\dot{y}/y \approx g - (1-\alpha)(\delta + g + n)(\log y - \log y^*)$, which is (3.11): the book's equation is the linearization of (*).

The rate of convergence is not a constant. Write $z \equiv \log y(t) - \log y^*(t)$ for the gap and define the convergence coefficient implicitly by $\dot{y}/y = g - \beta(z)z$, so that from (*)

$$\beta(z) = \alpha(\delta + g + n) \frac{1 - e^{-cz}}{z}, \quad c \equiv \frac{1-\alpha}{\alpha} > 0.$$

The function $\varphi(u) \equiv (1 - e^{-u})/u$ has $\varphi'(u) = e^{-u}[u + 1 - e^u]/u^2 < 0$ for $u \neq 0$, because $e^u > 1 + u$; and $\varphi(u) \rightarrow 1$ as $u \rightarrow 0$. Hence

$$\beta \text{ is strictly decreasing in } z, \quad \beta(0) = \alpha(\delta + g + n)c = (1-\alpha)(\delta + g + n),$$

$$\beta(z) \rightarrow \infty \text{ as } z \rightarrow -\infty, \quad \beta(z) \rightarrow 0 \text{ as } z \rightarrow +\infty.$$

So for a country *below* its steady state ($z < 0$, the empirically relevant case) the true speed of convergence *exceeds* the constant $(1-\alpha)(\delta + g + n)$ of the linearized equation, and it does so by more the poorer the country is; for a country above its steady state convergence is slower than the

linear approximation says. With the parameter values of EXAMPLE 3.1 ($\alpha = 1/3$, $\delta + g + n = 0.08$, so $(1 - \alpha)(\delta + g + n) = 0.0533$):

y/y^*	0.10	0.25	0.50	0.80	1.25	2.00	10.0
β	1.1465	0.2885	0.1154	0.0672	0.0430	0.0289	0.0115
$\beta/\beta(0)$	21.5	5.41	2.16	1.26	0.81	0.54	0.21

A country at a tenth of its steady-state income converges twenty times faster than the log-linear coefficient suggests. Figure 3 plots β .

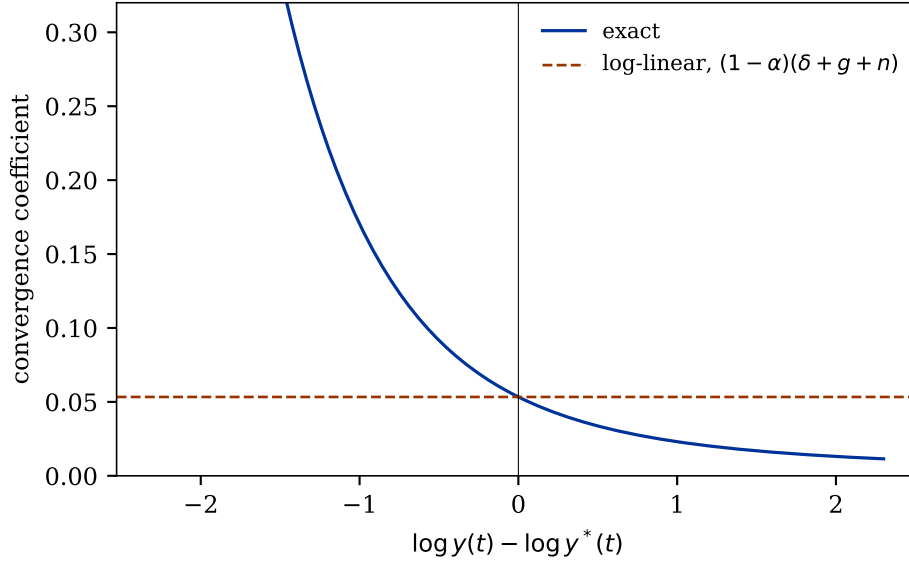


FIGURE 3. The exact convergence coefficient β as a function of the gap $\log y - \log y^*$, against the constant coefficient of the log-linearized equation (3.11).

Remark. This is a first hint of why growth regressions of the form (3.12)–(3.14), which impose a *constant* coefficient b_1 on $\log y_{i,t-1}$, are hard to interpret. Even within the Solow model with a common technology, the correct coefficient varies systematically with how far a country is from its own steady state, so the estimated b_1 is some average of very different speeds — and the average depends on the composition of the sample.

Exercise 3.4

Log-linear answer. By (3.11), each country's gap $z_i(t) \equiv \log y_i(t) - \log y^*(t)$ obeys $\dot{z}_i = -\beta z_i$ with $\beta = (1 - \alpha)(\delta + g + n)$. Since the two countries share $y^*(t)$, the *relative* gap $D(t) \equiv \log y_1(t) - \log y_2(t) = z_1(t) - z_2(t)$ obeys the same linear equation,

$$\dot{D} = -\beta D \quad \implies \quad D(t) = D(0)e^{-\beta t}.$$

With $\alpha = 1/3$, $g = 0.02$, $n = 0.01$ and $\delta = 0.05$ we get $\beta = \frac{2}{3} \times 0.08 = 0.0533$ per year. Starting from $D(0) = \log 2 = 0.6931$ and requiring $D(T) = \log 1.1 = 0.0953$,

$$T = \frac{1}{\beta} \log \frac{\log 2}{\log 1.1} = \frac{\log(7.273)}{0.0533} = \frac{1.9843}{0.0533} \approx 37.2 \text{ years.}$$

The half-life of the gap is $\log 2/\beta = 13.0$ years, which is the “little more than 10 years” quoted in EXAMPLE 3.1.

Exact answer. The Cobb–Douglas model can be solved in closed form, so the approximation can be checked. With $x \equiv k^{1-\alpha}$ the law of motion is linear, $\dot{x} = (1 - \alpha)[s - (\delta + g + n)x]$, so

$$\frac{x(t)}{x^*} = 1 + \left(\frac{x(0)}{x^*} - 1\right)e^{-\beta t}, \quad \frac{y(t)}{y^*(t)} = \left(\frac{x(t)}{x^*}\right)^{\frac{\alpha}{1-\alpha}},$$

with the *same* $\beta = (1 - \alpha)(\delta + g + n)$ now appearing exactly. The log gap between the two countries is no longer exactly exponential, because the map from x to y is concave, so the answer depends on where the pair sits relative to the common steady state:

country 1 starts at $y_1(0)/y^*$	0.8	1.0	2.0
years until $y_1/y_2 = 1.1$	21.4	27.4	49.9

The log-linear figure of 37 years is of the right order but not accurate: when both countries are below the steady state the true convergence is faster (Exercise 3.3), and when both are above it is slower.

Either way the message is the one the book draws: the basic Solow model with a capital share of one third predicts that a twofold income gap between otherwise identical countries is largely gone within a few decades. That is far faster than anything in the data — the gaps documented in Chapter 1 have persisted for a century and more — which is one of the motivations for the augmented model of Section 3.3 (where a larger effective capital share slows convergence) and, ultimately, for models in which y^* itself differs permanently across countries.

Exercise 3.5

Index economies by i and let each be a Solow economy with its own saving rate s_i , depreciation rate δ_i , population growth rate n_i and technology level $A_i(t)$, with a common rate of technological progress g (so that a stable world income distribution is possible). Each economy in isolation obeys (3.7), so its steady-state effective capital-labor ratio k_i^* solves

$$s_i \frac{f(k_i^*)}{k_i^*} = \delta_i + g + n_i,$$

and its steady-state output per capita is $y_i^*(t) = A_i(t)f(k_i^*)$.

Repeating the derivation of (3.10) economy by economy — nothing in it used cross-country homogeneity — gives the country-specific convergence equation

$$\frac{\dot{y}_i(t)}{y_i(t)} \approx g - (1 - \varepsilon_k(k_i^*))(\delta_i + g + n_i)(\log y_i(t) - \log y_i^*(t)).$$

Writing $\beta_i \equiv (1 - \varepsilon_k(k_i^*))(\delta_i + g + n_i)$ and discretizing between $t - 1$ and t ,

$$g_{i,t,t-1} \approx \underbrace{[g + \beta_i \log y_i^*(t - 1)]}_{b_i^0} - \beta_i \log y_{i,t-1}. \quad (*)$$

This is (3.13), $g_{i,t,t-1} = b_i^0 + b_1 \log y_{i,t-1} + \varepsilon_{i,t}$, with $b_1 = -\beta_i$: the constant term is country specific, precisely because the steady state y_i^* is. Two observations complete the derivation.

The slope is common when the elasticity is. With a Cobb–Douglas technology $\varepsilon_k \equiv \alpha$ and $\beta_i = (1 - \alpha)(\delta_i + g + n_i)$, so the slope varies across countries only through $\delta_i + n_i$. Since $\delta + n$ moves within a narrow range across countries while y_i^* varies enormously, treating b_1 as common and b_i^0 as country specific — exactly what (3.13) does — is the natural first approximation. (That b_1 is in principle country specific is the caveat the book records after (3.13).)

The intercept is a function of observable country characteristics. With Cobb–Douglas technology $y_i^*(t) = A_i(t)[s_i/(\delta_i + g + n_i)]^{\alpha/(1-\alpha)}$, so

$$b_i^0 = g + (1 - \alpha)(\delta_i + g + n_i) \left[\log A_i(t-1) + \frac{\alpha}{1-\alpha} (\log s_i - \log(\delta_i + g + n_i)) \right].$$

Modeling b_i^0 as a linear function of $\log s_i$, $\log(\delta_i + g + n_i)$ and proxies for $\log A_i$ — schooling, institutions, openness, and the rest — delivers the empirical specification (3.14), $g_{i,t-1} = X_{i,t}^T \beta + b_1 \log y_{i,t-1} + \varepsilon_{i,t}$. Conditional convergence is then the statement that $b_1 < 0$ once these characteristics are held fixed, whereas unconditional convergence — the regression (3.12) with a common intercept — need not show $b_1 < 0$ at all: omitting b_i^0 puts it in the error term, and since economies with low y_i^* also tend to have low $y_{i,t-1}$, the resulting $\text{Cov}(b_i^0, \log y_{i,t-1}) > 0$ biases the estimate of b_1 upward.

Exercise 3.6

The system to be studied is

$$\dot{k} = G_k(k, h) \equiv s_k f(k, h) - (\delta_k + g + n)k, \quad \dot{h} = G_h(k, h) \equiv s_h f(k, h) - (\delta_h + g + n)h,$$

on $\mathbb{R}_{++}^2$, where $f(k, h) = F(k, h, 1)$ is strictly increasing and strictly concave in (k, h) by ASSUMPTION 1' and satisfies the Inada conditions of ASSUMPTION 2'. By PROPOSITION 3.1 there is exactly one rest point $(k^*, h^*) \in \mathbb{R}_{++}^2$. We must show that every solution starting in $\mathbb{R}_{++}^2$ converges to it.

Step 1: the system is cooperative, hence the flow is monotone. The off-diagonal entries of the Jacobian are

$$\frac{\partial G_k}{\partial h} = s_k f_h(k, h) > 0, \quad \frac{\partial G_h}{\partial k} = s_h f_k(k, h) > 0,$$

so the vector field is *cooperative*: each variable helps the other accumulate — more human capital raises output and therefore physical investment, and conversely. By Kamke's comparison theorem the flow preserves the componentwise order: if $(k_1(0), h_1(0)) \leq (k_2(0), h_2(0))$ then $(k_1(t), h_1(t)) \leq (k_2(t), h_2(t))$ for all $t \geq 0$. Two consequences will be used repeatedly. First, a solution that starts above (below) (k^*, h^*) stays above (below) it, since (k^*, h^*) is itself a solution. Second, if the vector field is nonpositive in both coordinates at the initial point, then the solution is nonincreasing in both coordinates for all t : the solution started at time $\tau > 0$ lies below the one started at time 0, and both solve the same autonomous equation.

Step 2: a supersolution and a subsolution. Recall from the proof of PROPOSITION 3.1 that the loci $\{G_k = 0\}$ and $\{G_h = 0\}$ are increasing curves in (k, h) space that cross exactly once, at (k^*, h^*) , with $\{G_k = 0\}$ steeper there; moreover $\{G_k = 0\}$ lies below $\{G_h = 0\}$ for small (k, h) and above it for large (k, h) . Since G_k is increasing in h and G_h is increasing in k ,

$$G_k < 0 \text{ to the right of } \{G_k = 0\}, \quad G_h < 0 \text{ above } \{G_h = 0\}.$$

For $k > k^*$ the region between the two curves is therefore a set on which $G_k < 0$ and $G_h < 0$; it is nonempty and, because both curves are increasing and unbounded (by ASSUMPTION 2', $f(k, h)/k \rightarrow 0$ as $k \rightarrow \infty$ forces $h \rightarrow \infty$ along $\{G_k = 0\}$), it contains points $(\bar{k}, \bar{h})$ with $\bar{k} \geq k(0)$ and $\bar{h} \geq h(0)$, as large as we please. Symmetrically, for $k < k^*$ the region between the two curves has $G_k > 0$ and $G_h > 0$ and contains points $(\underline{k}, \underline{h})$ with $\underline{k} \leq k(0)$, $\underline{h} \leq h(0)$, as small as we please.

Step 3: the sandwich. Consider the solution started at $(\bar{k}, \bar{h})$. By Step 1 it is nonincreasing in both coordinates, and it is bounded below by (k^*, h^*) — it started above the rest point, so it stays above. A monotone bounded trajectory converges, and its limit is a rest point of the system; by uniqueness, the limit is (k^*, h^*) . Exactly the same argument applied at $(\underline{k}, \underline{h})$ gives a trajectory that is nondecreasing, bounded above by (k^*, h^*) , and convergent to it. Finally, by monotonicity of the flow, the solution started at $(k(0), h(0))$ is trapped between these two for all $t \geq 0$:

$$(\underline{k}(t), \underline{h}(t)) \leq (k(t), h(t)) \leq (\bar{k}(t), \bar{h}(t)),$$

and both bounds converge to (k^*, h^*) . Hence $(k(t), h(t)) \rightarrow (k^*, h^*)$ from any initial condition in $\mathbb{R}_{++}^2$, which is the assertion of PROPOSITION 3.2.

Remark. The phase diagram in Figure 3.1 of the book is exactly this argument in pictures: the two loci cut the positive quadrant into four regions, in two of which both variables move toward the steady state, while in the other two the trajectory must cross into one of those. What the monotone-flow argument adds is the step the picture cannot supply — that no trajectory can cycle around the rest point instead of converging to it. Cooperativity is what rules that out; for planar cooperative systems there are no closed orbits.

Exercise 3.7

The steady state solves (3.17)–(3.18),

$$\Phi_1(k, h) \equiv s_k f(k, h) - (\delta_k + g + n)k = 0, \quad \Phi_2(k, h) \equiv s_h f(k, h) - (\delta_h + g + n)h = 0.$$

Write J for the Jacobian of (Φ_1, Φ_2) in (k, h) , evaluated at (k^*, h^*) :

$$J = \begin{pmatrix} s_k f_k - (\delta_k + g + n) & s_k f_h \\ s_h f_k & s_h f_h - (\delta_h + g + n) \end{pmatrix} = \begin{pmatrix} J_{11} & J_{12} \\ J_{21} & J_{22} \end{pmatrix}.$$

Signs of the entries and of $\det J$. The off-diagonal entries are positive. For the diagonal ones, substitute $\delta_k + g + n = s_k f(k^*, h^*)/k^*$ from (3.17):

$$J_{11} = s_k \left[f_k(k^*, h^*) - \frac{f(k^*, h^*)}{k^*} \right] < 0,$$

because strict concavity of f in k together with $f(0, h^*) \geq 0$ gives $f > k f_k$; similarly $J_{22} < 0$. For the determinant, substituting both steady-state conditions,

$$\det J = (\delta_k + g + n)(\delta_h + g + n) - s_k f_k(\delta_h + g + n) - s_h f_h(\delta_k + g + n) = \frac{s_k s_h f}{k^* h^*} [f - k^* f_k - h^* f_h] > 0,$$

the bracket being positive by strict concavity of f (this is the same inequality that made the two loci cross only once in PROPOSITION 3.1).

The comparative statics. By the Implicit Function Theorem, $\partial(k^*, h^*)^T / \partial\theta = -J^{-1} \partial(\Phi_1, \Phi_2)^T / \partial\theta$ with

$$J^{-1} = \frac{1}{\det J} \begin{pmatrix} J_{22} & -J_{12} \\ -J_{21} & J_{11} \end{pmatrix}.$$

Since $\partial\Phi/\partial s_k = (f, 0)^T$, $\partial\Phi/\partial s_h = (0, f)^T$ and $\partial\Phi/\partial n = (-k^*, -h^*)^T$,

$$\begin{aligned} \frac{\partial k^*}{\partial s_k} &= \frac{-J_{22}f}{\det J} > 0, & \frac{\partial h^*}{\partial s_k} &= \frac{J_{21}f}{\det J} = \frac{s_h f_k f}{\det J} > 0, \\ \frac{\partial k^*}{\partial s_h} &= \frac{J_{12}f}{\det J} = \frac{s_k f_h f}{\det J} > 0, & \frac{\partial h^*}{\partial s_h} &= \frac{-J_{11}f}{\det J} > 0, \\ \frac{\partial k^*}{\partial n} &= \frac{J_{22}k^* - J_{12}h^*}{\det J} < 0, & \frac{\partial h^*}{\partial n} &= \frac{J_{11}h^* - J_{21}k^*}{\det J} < 0, \end{aligned}$$

the last two because $J_{11}, J_{22} < 0$ while $J_{12}, J_{21} > 0$. Since f is increasing in both arguments, output per effective unit of labor $\hat{y}^* = f(k^*, h^*)$ inherits the signs: it rises with either saving rate and falls with population growth.

Interpretation. The novelty relative to the basic model is the *cross* effect: a higher saving rate in physical capital raises the steady-state stock of *human* capital as well, and vice versa. The mechanism is the complementarity of the two factors in f — more physical capital raises output, and since a constant fraction s_h of output is invested in schooling, it raises human capital investment too. This is the “multiplier” that makes the augmented model fit cross-country income differences so much better than the basic one: the elasticity of steady-state income with respect to s_k is $\alpha/(1 - \alpha - \beta)$ rather than $\alpha/(1 - \alpha)$, as (3.22)–(3.23) show explicitly for the Cobb–Douglas case.

Exercise 3.8

The accounting equation. With $Y = F(K, H, AL)$, differentiate with respect to time and divide by Y :

$$\frac{\dot{Y}}{Y} = \frac{F_K K}{Y} \frac{\dot{K}}{K} + \frac{F_H H}{Y} \frac{\dot{H}}{H} + \frac{F_{AL} AL}{Y} \left(\frac{\dot{A}}{A} + \frac{\dot{L}}{L} \right),$$

where F_{AL} is the derivative with respect to the third argument. With competitive factor markets the elasticities are the observed income shares: $R = F_K$ gives $\alpha_K = F_K K/Y$; the rental rate of human capital $R_H = F_H$ gives $\alpha_H = F_H H/Y$; and the wage per worker is $w = AF_{AL}$, so $\alpha_L = wL/Y = F_{AL} AL/Y$. Constant returns in the three arguments make the three shares sum to one. Writing g_X for the growth rate of X ,

$$g_Y = \alpha_K g_K + \alpha_H g_H + \alpha_L (g_A + g_L)$$

so that the estimate of TFP growth — the part of output growth not accounted for by measured inputs — is

$$\hat{x} = g_Y - \alpha_K g_K - \alpha_H g_H - \alpha_L g_L, \quad \text{and} \quad g_A = \frac{\hat{x}}{\alpha_L}.$$

Two differences from the fundamental growth accounting equation (3.3) of the basic model deserve emphasis. First, the accumulation of human capital is now an *input*, so growth in schooling and skills is no longer part of the residual. Second, because technology is labor augmenting, the residual $\hat{x}$ measures $\alpha_L g_A$, not g_A : with a labor share of about 2/3, the implied rate of technological progress is one and a half times the measured residual.

Using data. One needs (i) output growth g_Y , from national accounts; (ii) capital stock growth g_K , from a perpetual inventory calculation on investment series; (iii) growth in efficiency units of labor g_H , constructed as in Section 3.5 from schooling data and Mincerian returns, and g_L from employment; and (iv) the three factor shares. The last item is the binding constraint, and it is where the exercise bites: national accounts report a split of value added into capital income and labor income, but labor income here is the sum of two conceptually different payments, $R_H H$ and wL . Some identifying assumption is unavoidable. The two used in practice are

- impose a functional form and calibrate: with $Y = K^\alpha H^\beta (AL)^{1-\alpha-\beta}$ as in (3.21), the shares are the constants α , β and $1 - \alpha - \beta$, and β is pinned down from microeconomic (Mincerian) estimates of the return to schooling; or
- avoid the split altogether by folding human capital into effective labor, $Y = F(K, AH)$, as in (3.29): then only two shares are needed, α_K and $1 - \alpha_K$, and the accounting equation collapses to $g_Y = \alpha_K g_K + (1 - \alpha_K)(g_A + g_H)$, at the price of imposing that raw labor and human capital are perfect substitutes in efficiency units.

Either way the residual absorbs everything the measurement misses — quality of schooling, capacity utilization, reallocation across sectors and firms, and pure measurement error — so it remains, in Abramovitz’s phrase, a measure of our ignorance; adding human capital narrows that ignorance rather than eliminating it.

Exercise 3.11

(a) *The accounting.* Work in per worker terms, $y_j \equiv Y_j/L_j$ and $\kappa_j \equiv K_j/L_j$. Constant returns make output per worker a function of the capital-labor ratio and technology alone, $y = \phi(\kappa; A)$, and competitive markets make the elasticity of that function equal to the *observed* capital share:

$$\frac{\partial \log y}{\partial \log \kappa} = \frac{F_K K}{Y} = \alpha^K.$$

This is all that is needed: no functional form, no assumption about how A enters. Order the countries by their capital-labor ratio, $\kappa_{(1)} \leq \kappa_{(2)} \leq \dots \leq \kappa_{(J)}$. If two adjacent countries had the same technology, the whole of their income difference would have to come from capital deepening,

$$\Delta \log y_{\text{capital}} = \bar{\alpha}^K \Delta \log \kappa, \quad \bar{\alpha}^K = \frac{1}{2}(\alpha_{(i)}^K + \alpha_{(i+1)}^K),$$

a Divisia (Törnqvist) index that uses the average of the two observed shares over the interval. The residual is the estimated technology gap,

$$\log \frac{A_{(i+1)}}{A_{(i)}} = (\log y_{(i+1)} - \log y_{(i)}) - \bar{\alpha}^K (\log \kappa_{(i+1)} - \log \kappa_{(i)}),$$

and cumulating these differences along the ordered cross section gives every country’s productivity relative to a numeraire country (the United States, say). Equivalently, and in one step for any pair,

$$\log \frac{A_j}{A_{\text{US}}} = \log \frac{y_j}{y_{\text{US}}} - \bar{\alpha}^K \log \frac{\kappa_j}{\kappa_{\text{US}}}.$$

The contribution of technology to cross-country income differences is then measured by the dispersion of $\log A_j$ relative to the dispersion of $\log y_j$, and the “predicted” income of country j

under a common technology is $\log \hat{y}_j = \log y_{US} + \bar{\alpha}^K \log(\kappa_j/\kappa_{US})$. (With a common Cobb–Douglas technology $\alpha_j^K \equiv \alpha$ and this is exactly the calibration of Section 3.5.) Only if the shares vary across countries does the chaining matter, and then the index is exact only up to the usual path dependence, which is second order when the share varies slowly.

(b) *What do large A differences mean?* Not necessarily that countries have access to different production possibility sets. The residual A_j is defined as whatever makes the accounting identity hold, so it absorbs every reason why a country produces less from the *measured* inputs than the numeraire country does:

- unmeasured or mismeasured inputs — human capital (part (c)), land and natural resources, capital quality and capacity utilization, hours per worker;
- allocative efficiency — misallocation of capital and labor across firms and sectors, distortions, credit constraints, monopoly power, informality;
- organization and institutions — the efficiency with which given inputs are combined;
- genuinely different technical knowledge or its adoption.

Only the last is technology in the engineering sense. Since a country can, in principle, adopt known technologies, the finding of very large A gaps is better read as a statement that *something other than the accumulation of measured factors* accounts for most of cross-country income differences — which is precisely what motivates Chapter 4’s search for fundamental causes.

(c) *What data are needed with $F(K, H, A)$?* Efficiency units of labor must be measured: the schooling distribution of the workforce (attainment by level), an estimate of how schooling maps into productivity (Mincerian returns, ideally allowing the return to vary by level and country), experience profiles, and — if one wants α_H separately from α_L — the split of labor income into payments to raw labor and to skills. Data on schooling quality (test scores, expenditure per pupil, teacher quality) are needed to avoid the bias in part (d), as are hours worked if labor input is to be measured consistently.

(d) *Quality of schooling shows up as productivity.* Suppose the truth is $Y_j = K_j^\alpha (AH_j)^{1-\alpha}$ with a common A , and that the true stock of efficiency units is $H_j = Q_j \hat{H}_j$, where $\hat{H}_j$ is the measure constructed from years of schooling as in Section 3.5 and $Q_j > 0$ captures the quality of a year of schooling. The calibration of Section 3.5 computes productivity as the residual from the measured stock $\hat{H}_j$:

$$\hat{A}_j \equiv \left[\frac{Y_j}{K_j^\alpha \hat{H}_j^{1-\alpha}} \right]^{\frac{1}{1-\alpha}} = \left[\frac{K_j^\alpha (AQ_j \hat{H}_j)^{1-\alpha}}{K_j^\alpha \hat{H}_j^{1-\alpha}} \right]^{\frac{1}{1-\alpha}} = AQ_j.$$

So the estimated productivity of country j relative to the United States is

$$\frac{\hat{A}_j}{\hat{A}_{US}} = \frac{Q_j}{Q_{US}},$$

even though all countries share the same technology: *every* estimated productivity difference is schooling quality in disguise. The bias goes the empirically awkward way, too. Poor countries are generally thought to have lower Q , so $\hat{A}_j$ is understated for them, and the calibration attributes to

“technology” what is really an unmeasured input — which is exactly the criticism of the Hall–Jones and Klenow–Rodríguez calibrations that the quality-adjustment literature (Caselli, 2005) takes up.

Chapter 4

Fundamental Determinants of Differences in Economic Performance

The two exercises of Chapter 4 left unsolved in the instructor's manual: Exercises 4.1 and 4.2, both on the Simon–Kremer model of Section 4.2. The manual solves 4.3. Throughout, world output is $Y(t) = L(t)^\alpha (A(t)Z)^{1-\alpha}$ with Z normalized to 1, so that $Y = L^\alpha A^{1-\alpha}$, and population responds to output through (4.2), $L(t) = \phi Y(t)$.

Exercise 4.1

Population as a function of the stock of ideas. Substituting the production function into (4.2),

$$L = \phi Y = \phi L^\alpha A^{1-\alpha} \quad \implies \quad L^{1-\alpha} = \phi A^{1-\alpha} \quad \implies \quad L(t) = \phi^{\frac{1}{1-\alpha}} A(t). \quad (*)$$

The Malthusian channel thus ties population to the stock of knowledge one for one: any improvement in technology is absorbed by a proportional increase in the number of people it has to feed.

Equation (4.3). Each individual discovers ideas at the rate λ , so by (4.1) $\dot{A} = \lambda L$. Substituting (*),

$$\dot{A}(t) = \lambda \phi^{\frac{1}{1-\alpha}} A(t),$$

which is (4.3): the *level* of the flow of new ideas is proportional to the existing stock, because the population that generates them is.

Equation (4.4). This is a linear differential equation with constant coefficient; separating variables, $d \log A = \lambda \phi^{1/(1-\alpha)} dt$, so with the initial condition $A(0) > 0$,

$$A(t) = \exp(\lambda \phi^{1/(1-\alpha)} t) A(0),$$

which is (4.4). Technology grows exponentially at the constant rate $g \equiv \lambda \phi^{1/(1-\alpha)}$, and so do population, by (*), and world output, since

$$Y = L^\alpha A^{1-\alpha} = (\phi^{\frac{1}{1-\alpha}} A)^\alpha A^{1-\alpha} = \phi^{\frac{\alpha}{1-\alpha}} A.$$

Remark. Income per capita, however, does not grow at all: $y = Y/L = 1/\phi$ at every date, directly from (4.2). The model produces a growing *world* — more people, more ideas, more output — with living standards pinned down by the Malthusian relation between income and population. That is the sense in which economies of scale in population are a theory of world growth rather than of prosperity, and it is the observation that Exercise 4.3(a) formalizes.

Exercise 4.2

Derivation. Simon and Kremer replace (4.1) by the assumption that population determines the *proportional* rate of improvement,

$$\frac{\dot{A}(t)}{A(t)} = \lambda L(t).$$

The static relations are unchanged, so (*) of Exercise 4.1 still gives $L(t) = \phi^{1/(1-\alpha)} A(t)$ and therefore

$$\dot{A}(t) = \lambda \phi^{\frac{1}{1-\alpha}} A(t)^2.$$

This is separable: $dA/A^2 = \lambda \phi^{1/(1-\alpha)} dt$, and integrating from 0 to t ,

$$\frac{1}{A(0)} - \frac{1}{A(t)} = \lambda \phi^{\frac{1}{1-\alpha}} t \quad \implies \quad A(t) = \frac{1}{A(0)^{-1} - \lambda \phi^{1/(1-\alpha)} t},$$

which is (4.5).

How the behavior differs. Under (4.4) the growth rate of technology is the constant $\lambda \phi^{1/(1-\alpha)}$. Under (4.5) it is

$$\frac{\dot{A}(t)}{A(t)} = \lambda \phi^{\frac{1}{1-\alpha}} A(t) = \frac{\lambda \phi^{1/(1-\alpha)}}{A(0)^{-1} - \lambda \phi^{1/(1-\alpha)} t},$$

which is increasing in t : growth accelerates. Indeed the acceleration is violent. The denominator of (4.5) vanishes at

$$T^* = \frac{1}{\lambda \phi^{1/(1-\alpha)} A(0)} < \infty,$$

so technology, population and output all become infinite in *finite* time; the solution does not merely grow fast, it blows up. Starting from a small $A(0)$, T^* is large and the early path is nearly flat, which is what gives the model its appeal as a story of millennia of stagnation followed by a takeoff.

Why. The difference is in how an idea is assumed to interact with the stock of ideas already available. Under (4.1) a discovery adds a fixed increment to the pool: a person is worth λ units of A per unit of time regardless of how much is already known. Under the Simon–Kremer assumption a discovery raises the stock *proportionally*: each person contributes λ percentage points of growth per unit of time, so a discovery is worth more the more is already known — the “standing on the shoulders of giants” externality. Since the Malthusian channel makes population itself proportional to A , the second assumption makes the growth rate proportional to A , and a differential equation whose growth rate grows linearly in the state explodes in finite time.

Which is more plausible? Neither is comfortable, and they fail in different places.

Taken as a description of the very long run, the Simon–Kremer version is the better one. It is essentially the only simple mechanism that delivers the qualitative shape of the historical record — hundreds of millennia of near-stagnation followed by an acceleration — and Kremer’s evidence that the growth rate of world population has risen with the *level* of world population over a million years is direct support for the proportional specification.

Taken literally as a model, however, it cannot be right. Finite-time blow-up is not a prediction about any observable world, so the specification must fail at some scale; and the mechanism is a pure scale effect, so it predicts that larger populations grow faster — a prediction rejected when the units are countries (the European population was below that of Asia for two millennia, yet modern growth began in Europe), and rescued only by insisting that the relevant unit is the world as a whole.

The assumption in (4.1) has the more defensible microfoundation — independent discoveries arriving at a constant rate per person, which is exactly the probabilistic story the section tells — but it can only generate steady exponential growth, never a takeoff.

The binding weakness, though, is shared and lies elsewhere: it is (4.2). Both models force income per capita to equal $1/\phi$ at every date, so neither can speak to living standards at all, and the Malthusian link between income and population on which they rest broke down precisely when modern growth began, with the demographic transition reversing the sign of the relation between income and fertility. A mechanism that is silent about income per capita and about *where* growth started is, as the chapter concludes, at best a theory of world growth — not a fundamental cause of cross-country differences.

Chapter 5

Foundations of Neoclassical Growth

The six exercises of Chapter 5 left unsolved in the instructor's manual: Exercises 5.3–5.8. The manual solves 5.1, 5.2 and 5.9–5.14. Notation follows the book: $\mathcal{H}$ is the set of households, $p = (p_1, \dots, p_N)$ the price vector, w^h household h 's income, $v^h(p, w^h)$ its indirect utility, ω^h its endowment and θ_f^h its share in firm f .

Exercise 5.3

Each household $h \in \mathcal{H}$ has indirect utility of the Gorman form (5.3),

$$v^h(p, w^h) = a^h(p) + b(p)w^h,$$

with a common $b(\cdot)$, and a positive demand for every commodity. We must show that aggregate demand is the demand of a single household with indirect utility $v(p, w) = a(p) + b(p)w$, where $a(p) \equiv \int_{h \in \mathcal{H}} a^h(p) dh$ and $w \equiv \int_{h \in \mathcal{H}} w^h dh$.

Step 1: what homogeneity forces. Any indirect utility function is homogeneous of degree 0 in (p, w) , so for every $\lambda > 0$ and every w^h ,

$$a^h(\lambda p) + b(\lambda p)\lambda w^h = a^h(p) + b(p)w^h.$$

Both sides are affine in w^h , and the household's income can be varied freely, so the constant terms and the slopes must agree separately:

$$a^h(\lambda p) = a^h(p) \quad \text{and} \quad b(\lambda p) = \lambda^{-1}b(p),$$

that is, a^h is homogeneous of degree 0 and b of degree -1 . Since v^h is strictly increasing in income, $b(p) > 0$.

Step 2: individual demands. Assume a^h and b differentiable. By Roy's Identity, household h 's demand for commodity j is

$$x_j^h(p, w^h) = -\frac{\partial v^h / \partial p_j}{\partial v^h / \partial w^h} = -\frac{1}{b(p)} \frac{\partial a^h(p)}{\partial p_j} - \frac{1}{b(p)} \frac{\partial b(p)}{\partial p_j} w^h. \quad (*)$$

(The hypothesis that each household consumes a positive amount of every commodity is what makes the demands interior, so that Roy's Identity applies and no household sits at a corner where the envelope argument would fail.) Note the structure: the Engel curve of household h for commodity j is a straight line in w^h whose *slope*, $-b^{-1} \partial b / \partial p_j$, does not depend on h . All the household-specific information sits in the intercept.

Step 3: aggregation. Integrating $(*)$ over $\mathcal{H}$ and exchanging differentiation and integration,

$$\begin{aligned} x_j(p) &\equiv \int_{h \in \mathcal{H}} x_j^h(p, w^h) dh = -\frac{1}{b(p)} \frac{\partial}{\partial p_j} \int_{h \in \mathcal{H}} a^h(p) dh - \frac{1}{b(p)} \frac{\partial b(p)}{\partial p_j} \int_{h \in \mathcal{H}} w^h dh \\ &= -\frac{1}{b(p)} \frac{\partial a(p)}{\partial p_j} - \frac{1}{b(p)} \frac{\partial b(p)}{\partial p_j} w. \end{aligned}$$

Now consider the hypothetical household with indirect utility $v(p, w) = a(p) + b(p)w$. It is a legitimate indirect utility function: a is homogeneous of degree 0 and b of degree -1 by Step 1, so v is homogeneous of degree 0; it is increasing in w and decreasing in each p_j wherever the individual ones are. Applying Roy's Identity to it gives demands

$$-\frac{1}{b(p)} \frac{\partial a(p)}{\partial p_j} - \frac{1}{b(p)} \frac{\partial b(p)}{\partial p_j} w,$$

which is exactly $x_j(p)$. Hence aggregate demand is generated by the maximization problem of the single household with indirect utility $v(p, w)$: the economy admits a representative household, and THEOREM 5.2 is proved.

Step 4: the representative household is strong. Aggregate demand depends on the distribution of income only through the aggregate $w = \int w^h dh$, so any redistribution of income across households leaves it unchanged. The converse also holds, which is why the Gorman form is not merely sufficient but necessary: if two households h and h' had Engel curves of different slopes for commodity j , that is $b_h^{-1} \partial b_h / \partial p_j \neq b_{h'}^{-1} \partial b_{h'} / \partial p_j$, then transferring a unit of income from h' to h would change aggregate demand for j , and no single household's demand function could reproduce aggregate demand at all income distributions.

Remark. Nothing in the argument used $N < \infty$ beyond writing the derivatives coordinate by coordinate, and nothing used $\mathcal{H}$ finite: the integral is a Lebesgue integral and covers a continuum of households. What the theorem does *not* deliver is a *normative* representative household — the utility $v(p, w)$ reproduces aggregate demand but need not rank allocations the way any Pareto criterion would. That requires the additional structure of THEOREM 5.3, in particular $a^h(p) = a^h$ independent of p for the converse direction.

Exercise 5.4

Let the commodity space be indexed by $j \in J \equiv [0, 1]$ with the Lebesgue measure (nothing depends on the particular index set; $J = \mathbb{R}_+$ for a continuous-time horizon reads the same way). A consumption bundle is a measurable function $x : J \rightarrow \mathbb{R}$, and we take the commodity space to be $L^\infty(J)$ with prices in its predual $L^1(J)$, so that the value of a bundle is

$$p \cdot x = \int_J p(j)x(j) dj,$$

which is finite for $p \in L^1$, $x \in L^\infty$. (Any dual pair with a bilinear form of this kind will do; what matters is that values are given by integrals and that the dual space is rich enough to separate points.) Household h 's preferences have the Gorman representation $v^h(p, w^h) = a^h(p) + b(p)w^h$, each household demands a positive amount of (almost) every commodity, and the aggregate production possibilities set $Y \subset L^\infty(J)$ is convex. The claim is that THEOREM 5.3 survives verbatim:

- (1) any feasible allocation that maximizes $v(p, w) = \int_{\mathcal{H}} a^h(p) dh + b(p)w$, with $w = \int_{\mathcal{H}} w^h dh$, is Pareto optimal;
- (2) if in addition $a^h(p) = a^h$ for all p and all h , then every Pareto optimal allocation maximizes $v(p, w)$.

What carries over unchanged. The proof of THEOREM 5.3 is an argument about the *objective*

$$\max_{h \in \mathcal{H}} \sum \alpha^h v^h(p, w^h) = \max \left[\int_{\mathcal{H}} \alpha^h a^h(p) dh + b(p) \int_{\mathcal{H}} \alpha^h w^h dh \right]$$

subject to market clearing and to net supplies being profit maximizing at p , exactly as in (5.7). Every step of it is linear in the commodity index: sums $\sum_{j=1}^N p_j y_j$ become integrals $\int_J p(j) y(j) dj$, and Roy's Identity becomes the same formula with the functional (Gâteaux) derivatives $\delta a^h(p)/\delta p(j)$ and $\delta b(p)/\delta p(j)$ in place of partial derivatives. The linearity of the Gorman form in w^h — which is what makes the Pareto weights enter only through the weighted aggregate income and lets one substitute income transfers for commodity reallocations — is a statement about incomes, not about commodities, so it is untouched. With $b(p) > 0$, the objective is still increasing in aggregate income, and part 1 follows as before; with $a^h(p) = a^h$ constant, the argument by contradiction that produced (5.9)–(5.11) goes through with integrals.

What genuinely needs an extra assumption. Two things that are automatic in $\mathbb{R}^N$ are not automatic here.

- *Existence of supporting prices.* In $\mathbb{R}^N$ a convex set can always be separated from a disjoint convex set. In an infinite-dimensional space the separating hyperplane theorem requires a topological condition — for instance that Y (or the set of preferred allocations) have nonempty interior in the sup-norm topology, or that preferences satisfy Mas-Colell's properness condition. Without it a Pareto optimum need not be supported by any price system in the chosen dual space, and the phrase “at price vector p ” in the theorem would be empty. This is the same hypothesis the book adds in THEOREMS 5.6 and 5.7 when it proves the Second Welfare Theorem with infinitely many commodities; here it is needed for exactly the same reason.
- *Integrability.* The aggregates $a(p) = \int_{\mathcal{H}} a^h(p) dh$ and $w = \int_{\mathcal{H}} w^h dh$ must be finite, and the interchange of $\delta/\delta p(j)$ with $\int_{\mathcal{H}} dh$ must be legitimate — guaranteed, for instance, by a uniform integrable bound on the family $\{\delta a^h(p)/\delta p(j)\}$. With a continuum of commodities one must also require $p \cdot x^h < \infty$ for each household, which Exercise 5.5 delivers from local nonsatiation.

Conclusion. With these two provisos — a price system exists in the dual, and the household aggregates are well defined — THEOREM 5.3 holds word for word with $\sum_{j=1}^N$ replaced by $\int_J dj$. The economics of the result is therefore robust to the infinite-dimensional setting that growth models live in: it is the *linearity of Gorman preferences in income*, not the finiteness of the commodity space, that makes the representative household normative.

Exercise 5.5

“Household h chooses x^h at prices p ” means that x^h solves

$$\max_{x \in X^h} U^h(x) \quad \text{subject to} \quad p \cdot x \leq w^h, \quad w^h \equiv p \cdot \left(\omega^h + \sum_{f \in \mathcal{F}} \theta_f^h y^f \right),$$

and local nonsatiation is DEFINITION 5.3: at every $x \in X^h$, U^h is strictly increasing in at least one of its arguments, and $U^h(x) < \infty$.

Step 1: the income of a locally nonsatiated household that makes a choice is finite. Suppose $w^h = \infty$. Then the budget constraint does not bind anywhere and the problem reduces to maximizing U^h over X^h . But local nonsatiation says that at *every* point of X^h there is a coordinate j' in which U^h is strictly increasing, so for any candidate x the bundle $x(\varepsilon)$ that adds $\varepsilon > 0$ to coordinate j' — which is in X^h for small ε , this being exactly the construction used in the proof of THEOREM 5.5 — satisfies $U^h(x(\varepsilon)) > U^h(x)$ and is still affordable. No maximum exists, so no bundle is chosen, contradicting the hypothesis. Hence $w^h < \infty$.

Step 2: the chosen bundle is affordable, hence of finite value. Since x^h is in the budget set,

$$p \cdot x^h \leq w^h < \infty.$$

Step 3: in fact the budget holds with equality. Suppose $p \cdot x^h < w^h$. By local nonsatiation there is a coordinate j' in which U^h is strictly increasing at x^h ; for ε small enough the bundle $x^h(\varepsilon)$ obtained by adding ε to that coordinate still costs less than w^h and yields strictly higher utility, contradicting optimality. Therefore

$$p \cdot x^h = w^h < \infty.$$

Remark. In a finite-dimensional economy this is a triviality — a finite sum of finite terms. Its content is in the infinite-dimensional case relevant to growth theory, where $p \cdot x = \sum_{t=0}^{\infty} p_t x_t$ is an infinite series that may diverge. The result says that at any chosen bundle the series converges, and this is precisely what the proof of the First Welfare Theorem needs: (5.17)–(5.18) are summed over households, and the sums are meaningless unless each term is finite. It is also the reason DEFINITION 5.3 builds $U^h(x^h) < \infty$ into local nonsatiation: an infinite-utility plan would allow a household to be indifferent to budget constraints, and both the definition of a competitive equilibrium and the welfare theorems would lose their force.

Exercise 5.6

(a) *Demands and indirect utility.* Write $\hat{x}_j^h \equiv x_j^h - \xi_j^h$ and $\hat{w}^h \equiv w^h - \sum_j p_j \xi_j^h > 0$ (positive by the assumption $\sum_j p_j \bar{\xi} < w^h$). The problem

$$\max \left[\sum_{j=1}^N (\hat{x}_j^h)^{\frac{\sigma-1}{\sigma}} \right]^{\frac{\sigma}{\sigma-1}} \quad \text{s.t.} \quad \sum_{j=1}^N p_j \hat{x}_j^h \leq \hat{w}^h$$

is the textbook CES problem. The first-order conditions give $(\hat{x}_j^h)^{-1/\sigma} = \mu p_j$ for a multiplier $\mu > 0$, hence $\hat{x}_j^h \propto p_j^{-\sigma}$, and the budget constraint fixes the factor:

$$\hat{x}_j^h = \frac{\hat{w}^h p_j^{-\sigma}}{P}, \quad P \equiv \sum_{k=1}^N p_k^{1-\sigma}, \quad \text{that is} \quad x_j^h = \xi_j^h + \frac{(w^h - \sum_k p_k \xi_k^h) p_j^{-\sigma}}{\sum_k p_k^{1-\sigma}}.$$

Substituting back,

$$\sum_j (\hat{x}_j^h)^{\frac{\sigma-1}{\sigma}} = \left(\frac{\hat{w}^h}{P} \right)^{\frac{\sigma-1}{\sigma}} \sum_j p_j^{1-\sigma} = \left(\frac{\hat{w}^h}{P} \right)^{\frac{\sigma-1}{\sigma}} P,$$

so that

$$v^h(p, w^h) = \left[\left(\frac{\hat{w}^h}{P} \right)^{\frac{\sigma-1}{\sigma}} P \right]^{\frac{\sigma}{\sigma-1}} = \hat{w}^h P^{\frac{1}{\sigma-1}} = \frac{-\sum_j p_j \xi_j^h + w^h}{[\sum_j p_j^{1-\sigma}]^{\frac{1}{1-\sigma}}},$$

which is (5.5). It is of the Gorman form (5.3) with

$$b(p) = P^{\frac{1}{\sigma-1}} > 0 \quad \text{and} \quad a^h(p) = -\left(\sum_j p_j \xi_j^h\right) P^{\frac{1}{\sigma-1}},$$

and the household-specific term enters only through a^h , exactly as THEOREM 5.2 requires. It is homogeneous of degree 0: $\hat{w}^h$ is homogeneous of degree 1 in (p, w) and $P^{1/(\sigma-1)}$ of degree -1 .

(b) *The representative household.* The utility function (5.6) is the same CES form with $\xi_j \equiv \int_{\mathcal{H}} \xi_j^h dh$, so the computation in (a) applies verbatim and gives

$$v(p, w) = \frac{-\sum_j p_j \xi_j + w}{[\sum_j p_j^{1-\sigma}]^{\frac{1}{1-\sigma}}} = \int_{\mathcal{H}} a^h(p) dh + b(p)w,$$

using $\int_{\mathcal{H}} a^h(p) dh = -(\sum_j p_j \xi_j) P^{1/(\sigma-1)}$ and $w = \int_{\mathcal{H}} w^h dh$. This is precisely the indirect utility of the representative household constructed in THEOREM 5.2, so (5.6) is the utility function that rationalizes aggregate demand.

(c) *Dropping the outer exponent.* Let now $\tilde{U}^h = \sum_j (x_j^h - \xi_j^h)^{(\sigma-1)/\sigma}$. Take $\sigma > 1$, so that $(\sigma-1)/\sigma > 0$ and $\tilde{U}^h = (U^h)^{(\sigma-1)/\sigma}$ is a strictly increasing transformation of the utility in part (a). Demands are therefore unchanged, and the indirect utility function is the same transformation of (5.5):

$$\tilde{v}^h(p, w^h) = [\hat{w}^h P^{\frac{1}{\sigma-1}}]^{\frac{\sigma-1}{\sigma}} = (w^h - \sum_j p_j \xi_j^h)^{\frac{\sigma-1}{\sigma}} P^{\frac{1}{\sigma}}.$$

It is still homogeneous of degree 0 — the first factor has degree $(\sigma-1)/\sigma$ and the second $(1-\sigma)/\sigma$ — but it is *not* of the Gorman form: it is a strictly concave, not affine, function of w^h , so the Engel curves are no longer straight lines. The monotone transformation $z \mapsto z^{\sigma/(\sigma-1)}$ restores (5.5) and with it the Gorman form.

Is that enough? For the positive question — does the economy admit a (strong) representative household? — yes. Demand is an ordinal object: it is invariant to strictly increasing transformations of utility, so what matters is whether *some* representation of preferences takes the Gorman form, not whether the one we happened to write down does. This is the point the book makes right after THEOREM 5.2, and part (c) is the example that makes it concrete: aggregate demand here is exactly as in part (b).

Two caveats keep the answer from being an unqualified yes. First, under uncertainty utility is cardinal: the transformation $z \mapsto z^{\sigma/(\sigma-1)}$ changes the household's attitude to risk, so two representations that are equivalent for a deterministic exchange economy are *not* equivalent once expected utility is taken over lotteries — the aggregation result survives only for the representation that is also the Bernoulli utility index. Second, for a *normative* representative household the transformation matters as well: THEOREM 5.3(2) requires the Gorman representation to have $a^h(p) = a^h$ independent of prices, which is a restriction on a particular cardinalization, and Pareto weights attached to $\tilde{v}^h$ and to v^h define different social objectives. Aggregation of demand is ordinal; welfare analysis is not.

Exercise 5.7

Survival. Death arrives as the first event of a Poisson process with intensity ν , so the random date of death T is exponentially distributed:

$$\Pr\{T > t\} = e^{-\nu t}, \quad \mathbb{E}[T] = \frac{1}{\nu} < \infty.$$

The individual's expected lifetime is $1/\nu$, which is how one calibrates ν .

Expected utility. Normalize the flow utility after death to $u(0) = 0$, and let $[c(t)]_{t \geq 0}$ be the consumption plan conditional on being alive. Discounting the future at the pure rate ρ , expected lifetime utility at date 0 is

$$U_0 = \mathbb{E} \left[\int_0^T e^{-\rho t} u(c(t)) dt \right] = \mathbb{E} \left[\int_0^\infty e^{-\rho t} u(c(t)) \mathbf{1}\{T > t\} dt \right].$$

The integrand is nonnegative and measurable, so Fubini's theorem allows the expectation to be taken inside:

$$U_0 = \int_0^\infty e^{-\rho t} u(c(t)) \Pr\{T > t\} dt = \int_0^\infty e^{-\rho t} e^{-\nu t} u(c(t)) dt = \int_0^\infty e^{-(\rho+\nu)t} u(c(t)) dt.$$

This is the objective of an infinitely lived individual with discount rate

$\rho_{\text{eff}} = \rho + \nu$

Since the constraint set — the flow budget constraint and the no-Ponzi condition — is unaffected by the *form* of the objective, the individual's problem is identical to that of an infinitely lived agent who discounts at $\rho + \nu$. Her behavior, in particular her Euler equation $\dot{c}/c = (r(t) - \rho - \nu)/\theta$ for the CRRA case, is that of the infinite-horizon model with the larger discount rate.

Consistency with (5.12). In discrete time the book obtains $\beta = \hat{\beta}(1 - \nu)$; writing $\hat{\beta} = e^{-\rho\Delta}$ and $1 - \nu \approx e^{-\nu\Delta}$ for a period of length Δ gives $\beta \approx e^{-(\rho+\nu)\Delta}$, which is the same statement: the probability of not surviving acts exactly like impatience.

Remark. The result says that mortality raises the *effective discount rate*, not that it makes people poorer. Whether it also changes consumption growth depends on the asset market. With actuarially fair annuities — the Blanchard–Yaari setting of Chapter 9 — a surviving individual earns $r + \nu$ rather than r on her savings, because the assets of those who die are shared among the survivors, and the two effects cancel in the individual Euler equation: $\dot{c}/c = [(r + \nu) - (\rho + \nu)]/\theta = (r - \rho)/\theta$. What mortality then changes is the *aggregate*, through the arrival of newborns who hold no assets, which is exactly why the perpetual youth model breaks Ricardian equivalence while the individual problem looks infinite horizon.

Exercise 5.8

(a) *Different utility functions across generations.* Yes — the dynastic recursion still collapses into an infinite-horizon maximization, but the problem loses stationarity. Let generation t value its own consumption through u_t and the continuation value of its descendant with factor β :

$$V_t(y_t) = \max_{c_t + b_t \leq y_t} \{u_t(c_t) + \beta V_{t+1}(b_t + w)\}.$$

Iterating forward,

$$V_0 = \max \sum_{t=0}^{\infty} \beta^t u_t(c_t),$$

provided the sum converges (e.g. if the u_t are uniformly bounded). So a dynasty of one-period-lived altruistic individuals again behaves like a single infinitely lived agent.

Two things should be said about what does and does not survive. The objective remains *time-consistent*: viewed from date t , the dynasty maximizes $\sum_{s \geq t} \beta^{s-t} u_s(c_s)$, which ranks continuation plans exactly as $\sum_{s \geq 0} \beta^s u_s(c_s)$ does, because the two differ by the positive factor β^t and by terms that are already sunk. Geometric discounting, not the identity of the period utility functions, is what delivers consistency (compare EXERCISE 5.1). What is lost is *stationarity*: the maximization problem is no longer autonomous, so the stationary dynamic programming of Chapter 6 does not apply directly and one needs its nonstationary version (Section 6.7), with value functions indexed by time. Balanced growth also imposes a restriction: if $u_t(c) = \gamma^t u(c)$ for a constant $\gamma > 0$ — taste change at a constant rate — the objective is $\sum (\beta\gamma)^t u(c_t)$ and stationarity is restored with the modified discount factor $\beta\gamma$, which must satisfy $\beta\gamma < 1$. For general sequences $\{u_t\}$ neither a steady state nor a balanced growth path need exist.

(b) *Caring about grandchildren directly.* Now

$$V_t = u(c_t) + \beta V_{t+1} + \delta V_{t+2}, \quad \beta, \delta > 0.$$

Expanding, the weight a_s that generation t attaches to $u(c_{t+s})$ obeys the linear recursion

$$a_0 = 1, \quad a_1 = \beta, \quad a_s = \beta a_{s-1} + \delta a_{s-2} \quad (s \geq 2),$$

whose solution is a combination of the powers of the roots of $x^2 = \beta x + \delta$, the dominant one being $\varphi = (\beta + \sqrt{\beta^2 + 4\delta})/2$. Three consequences follow.

First, the objective is still an infinite discounted sum, $V_t = \sum_{s \geq 0} a_s u(c_{t+s})$, and it converges if and only if $\varphi < 1$, which is equivalent to

$$\beta + \delta < 1.$$

So the extra altruism must be paid for out of the total weight on the future.

Second — and this is the substantive difference — the discounting is *not* geometric. The ratios of successive weights are

$$\frac{a_1}{a_0} = \beta, \quad \frac{a_2}{a_1} = \beta + \frac{\delta}{\beta} > \beta,$$

and they keep changing (converging to φ only in the limit). A sequence of weights that falls at a varying rate is exactly the structure of the hyperbolic discounting of EXERCISE 5.1(b), and it has the same consequence: the plan that generation t would like its descendants to follow is not the plan that generation $t+1$ chooses, because generation t discounts dates $t+1$ and $t+2$ in the ratio β to $\beta^2 + \delta$, while generation $t+1$ discounts the same two dates in the ratio 1 to β . Preferences are *time-inconsistent*, and the dynasty can no longer be represented as a single maximizing agent.

Third, the modeling consequence: the right solution concept is not a maximization problem but an intergenerational *game*, in which each generation takes the decision rules of its descendants as given; the natural equilibrium notion is a Markov-perfect (or “consistent-planning”) equilibrium,

and saving behavior in it generally differs from the commitment solution — the dynasty saves less than generation t would wish, since it cannot bind its descendant to treat the grandchild as generously as it would like. The economics is the same as in the individual hyperbolic-discounting model: an extra, non-geometric concern for a more distant date creates a conflict between successive decision makers.